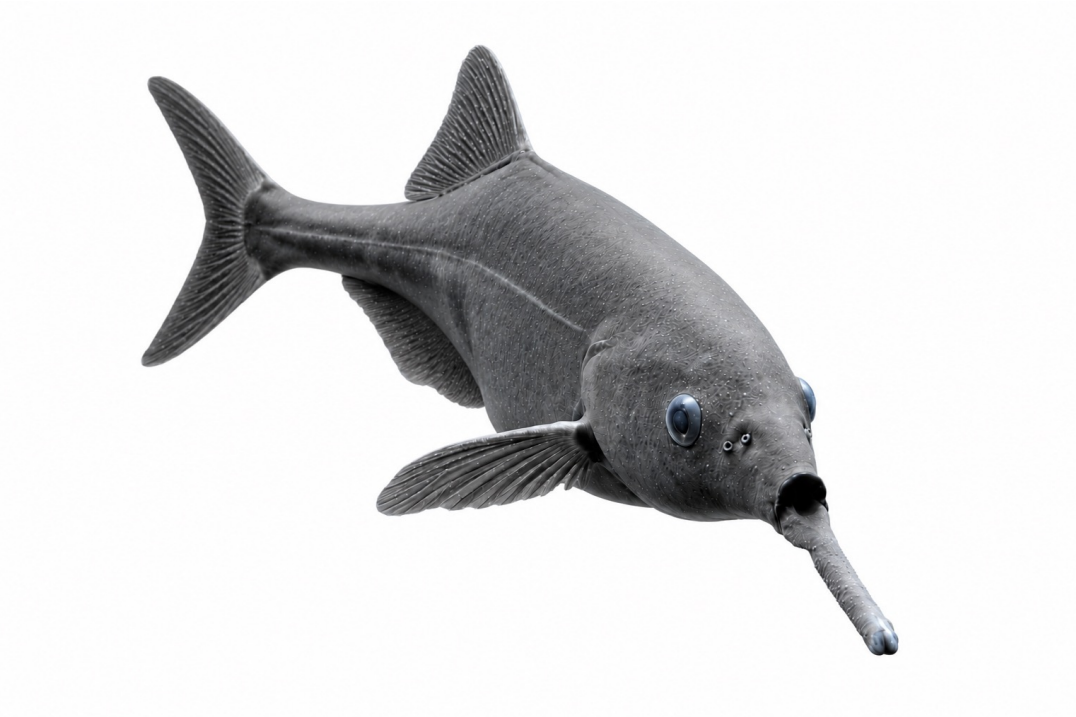
TECHNION
Israel Institute
of Technology

Sensory Systems: Electroreception

Avner Wallach

Laboratory of Sensing, Navigation, and Control in Living Systems
Faculty of Biology, Technion, Israel



Overview

- Introduction:
  - Electrogenesis and electroreception
  - Active electrorecption: weakly electric fish
- Electrolocation in pulse type fish
  - Adaptive filtering of self-generated noise
- Electrocommunication in wave type fish
  - Jamming Avoidance
  - Neural correlates of social cognition
- Learning and Navigation

Electrogenesis

Electrogenesis: the biological ability to generate electrical signals



Mastaba of Ti in Saqqara

Greek: *Narkē* (*narkan=to numb*)
Latin: *Torpedo* (*torpere=to paralyze*)



Electric Ray



John Walsh



John Hu





-400

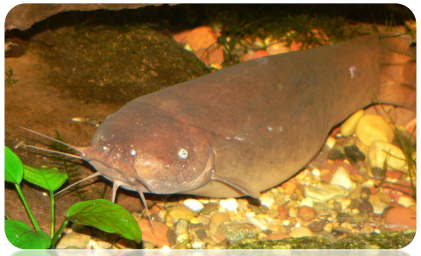
1800

-2750

BC 0 AD

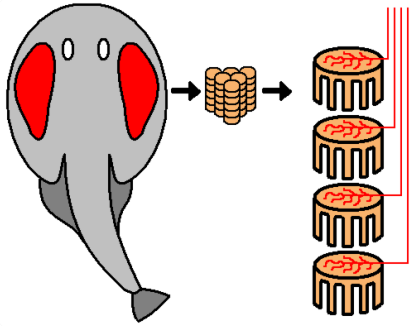
1773

Present



Electric Catfish









Alessandro Volta

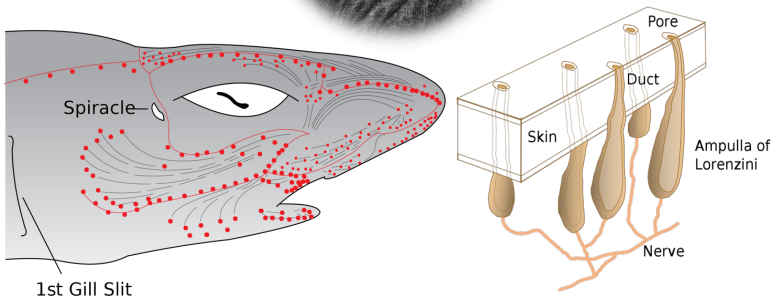
'Voltaic Pile'

Electroreception

Electroreception: the biological ability to perceive electrical signals

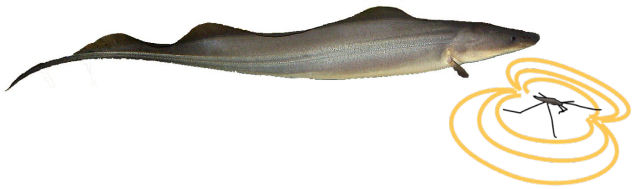
Stefano Lorenzini. 1678



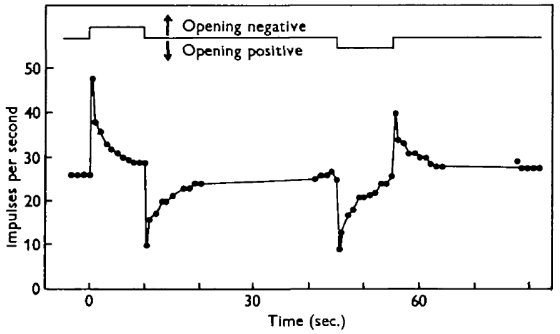


Hans Lissman, 1949

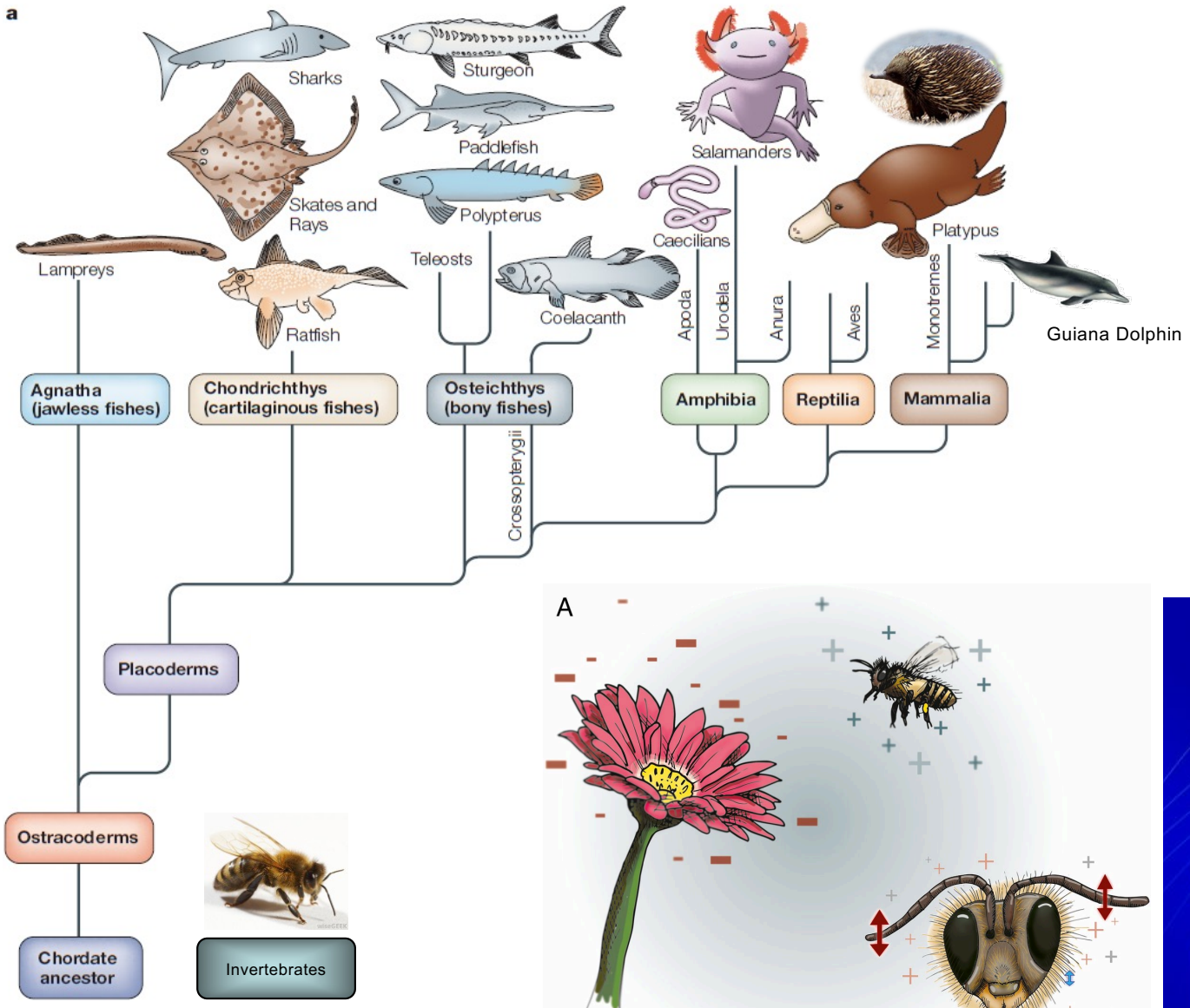




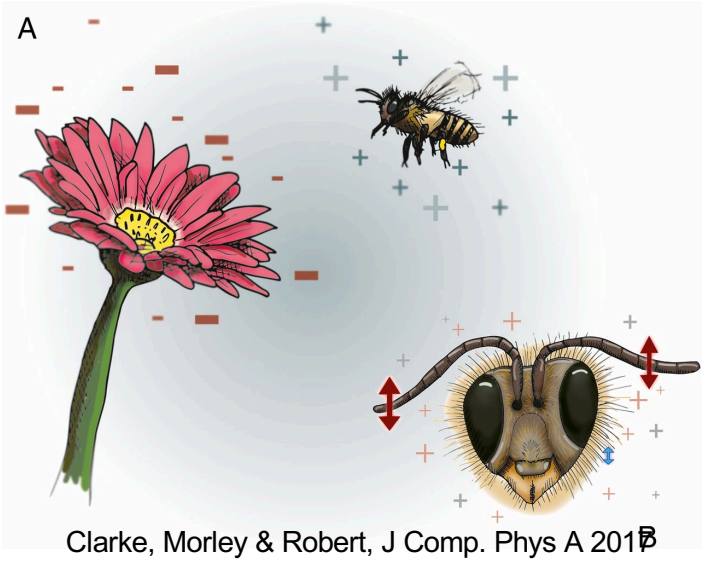
Richard W Murray, 1960



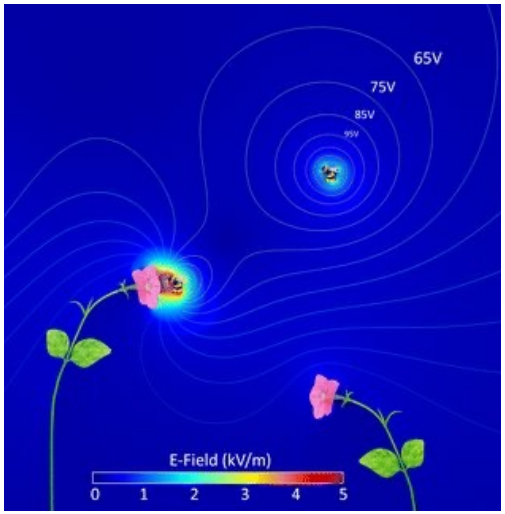
Electroreception in the Animal Kingdom



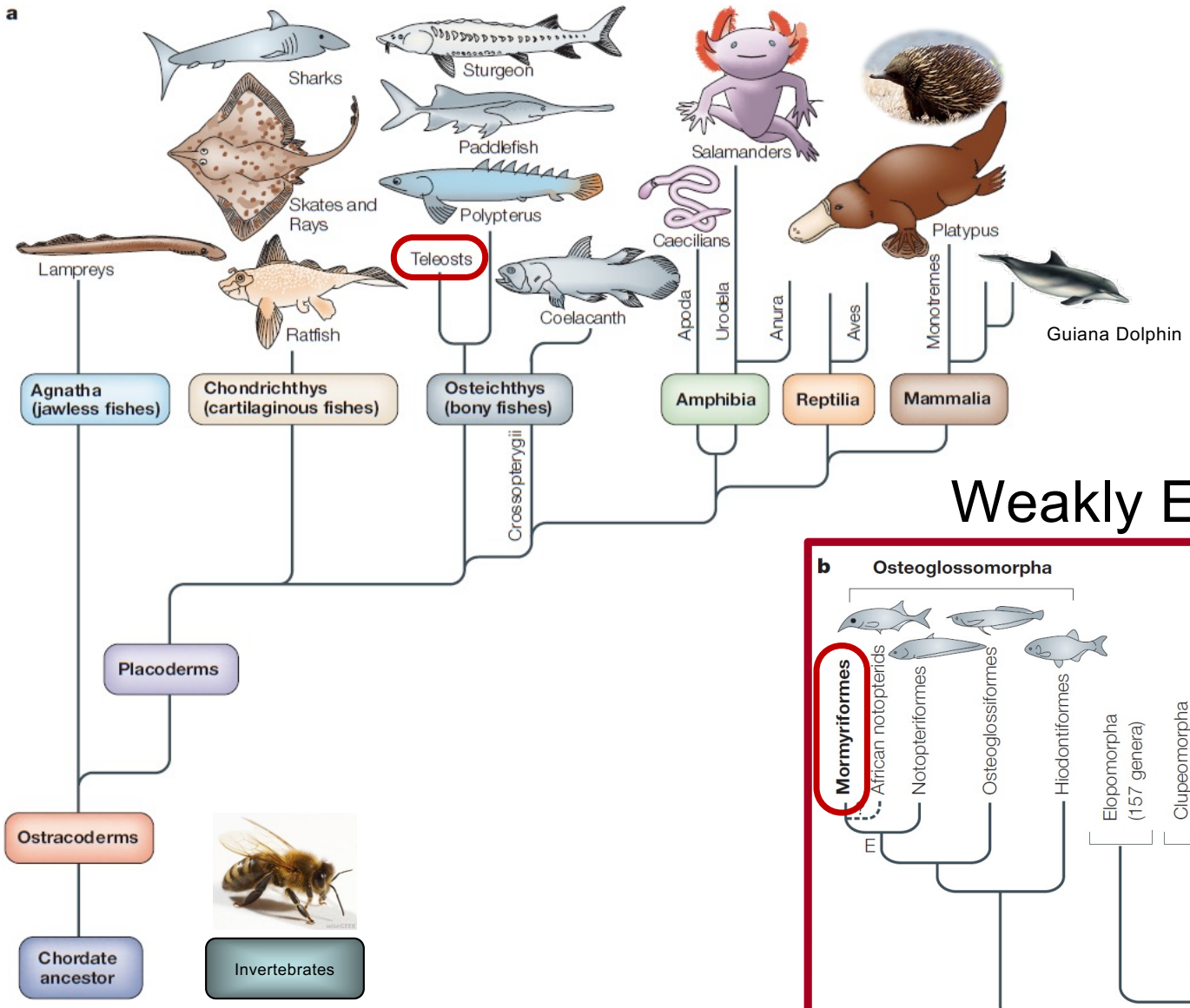
Rose, NRN 2004



Clarke, Morley & Robert, J Comp. Phys A 2017

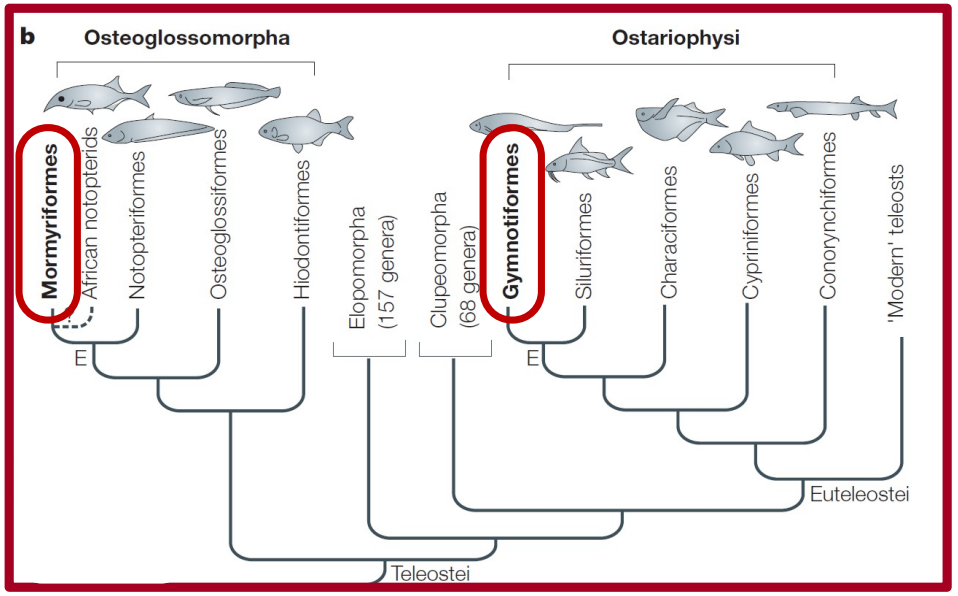


Electroreception in the Animal Kingdom

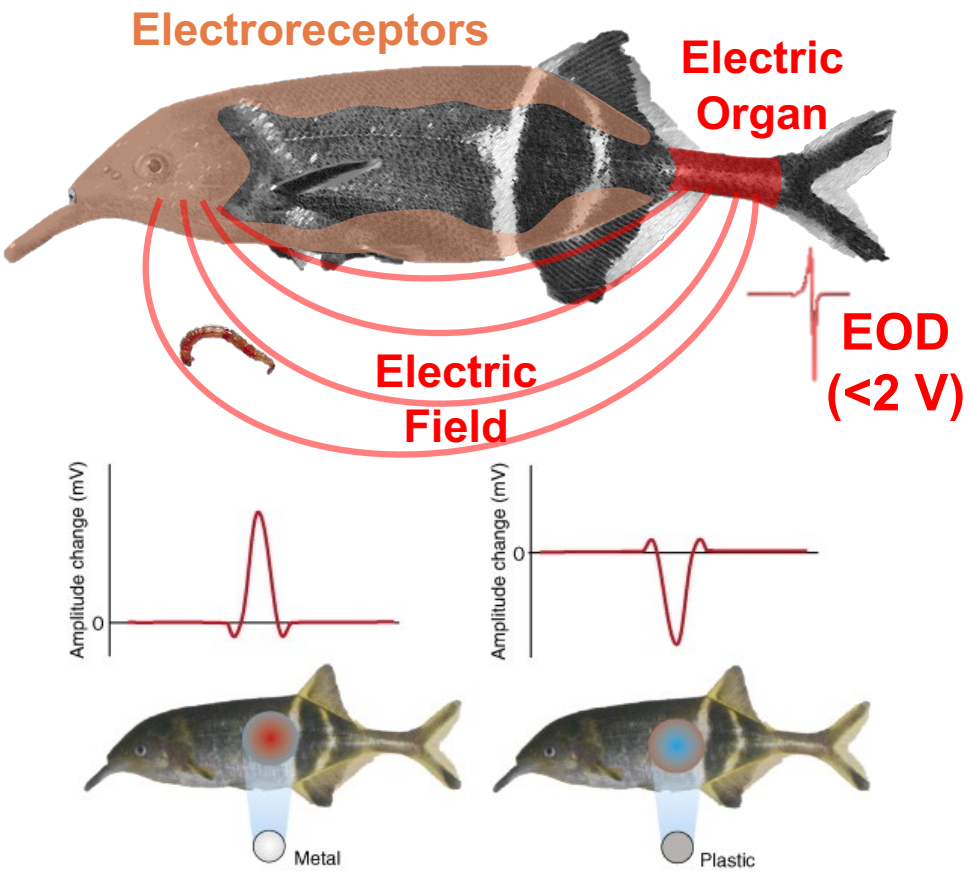


Rose, NRN 2004

Weakly Electric Fish



Active Electrorec

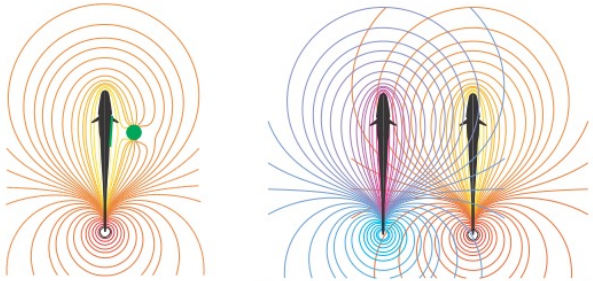


Short Range



Electrolocation and Electrocommunication

- Electrolocation: perception of external objects based on their 'electric image'
- Electrocommunication: interactions between the electric discharge of neighboring fish



Electrolocation **Electro-communication**

Octavolateral Systems

Active Electroreception

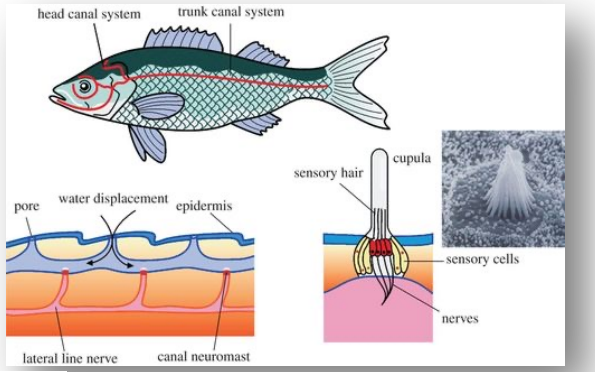
Cerebellum

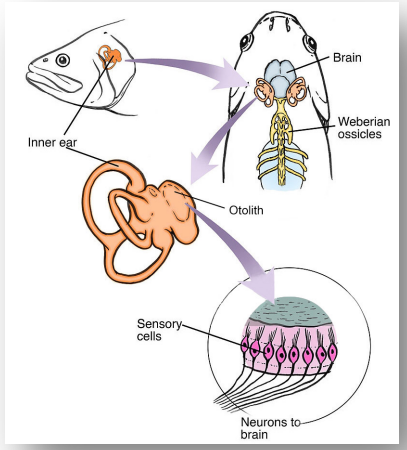
Auditory

Passive Electroreception

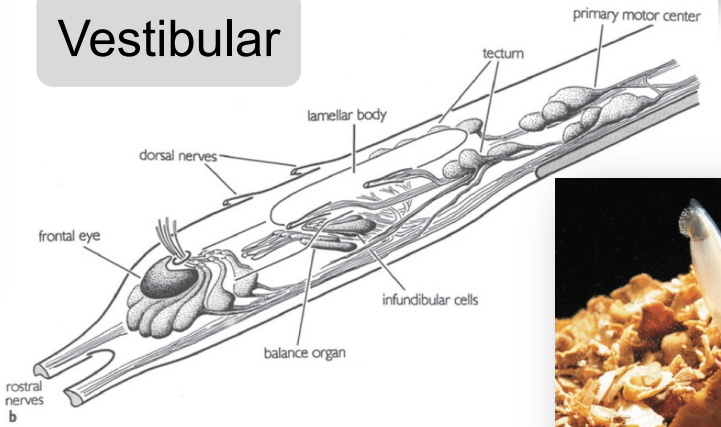
?

Lateral Line



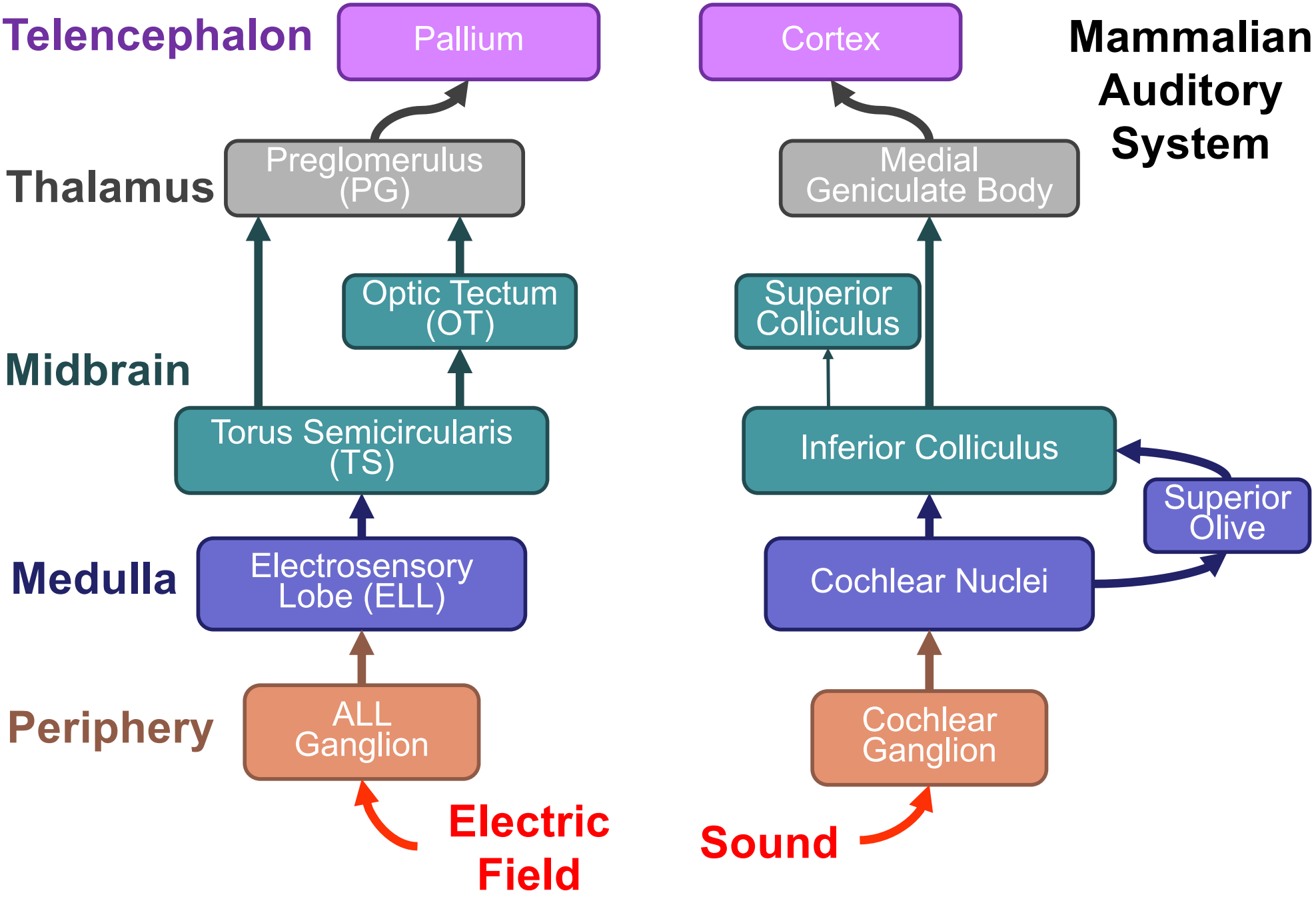


Vestibular



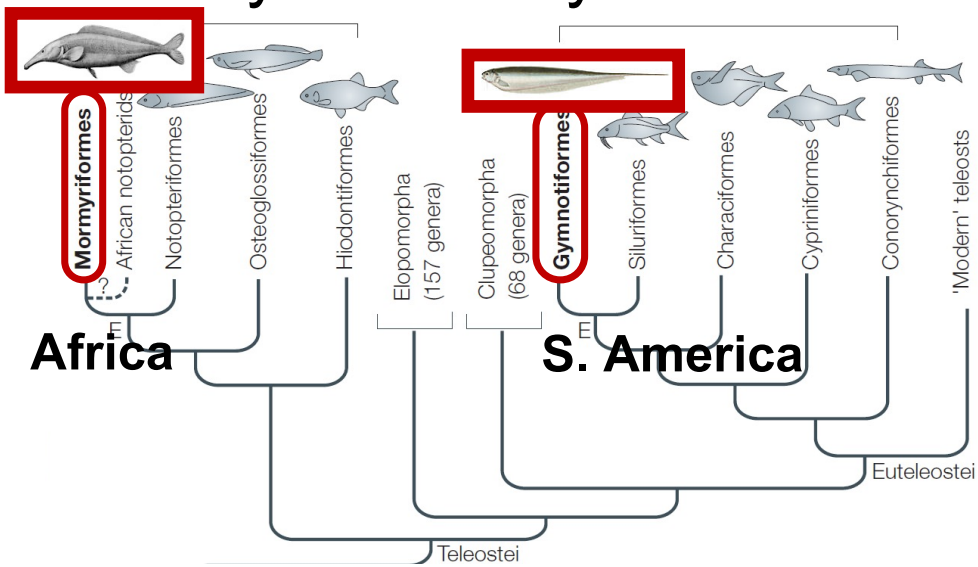


Ascending Electrosensory Pathway

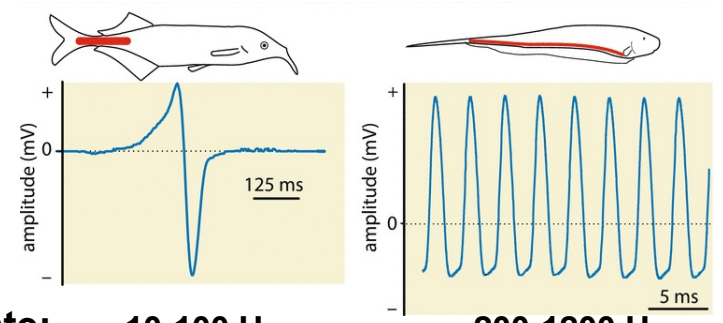


The Weakly Electric Fish

- Mormyrids vs. Gymnotids

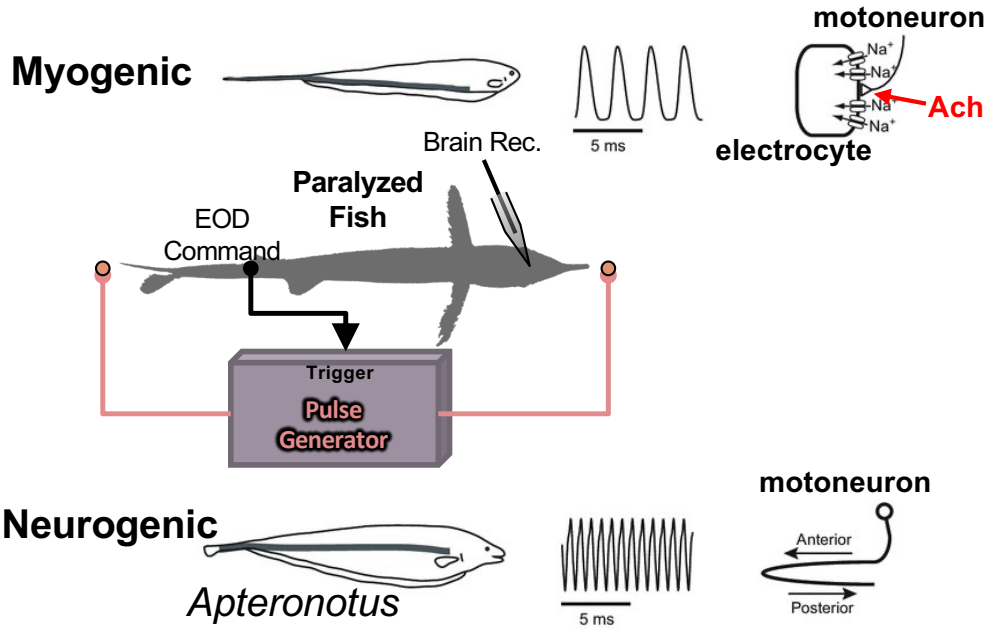


- Pulse vs. Wave

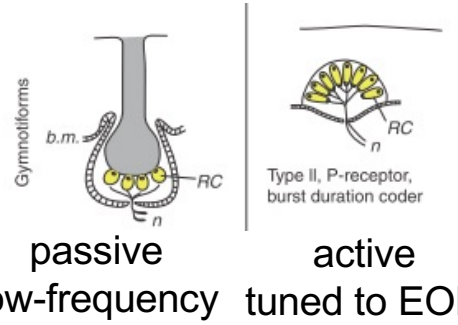


Rate:	**10-100 Hz**	**200-1200 Hz**
Rate Changes:	**Attention Communication**	**Communication**

- Myogenic vs. Neurogenic

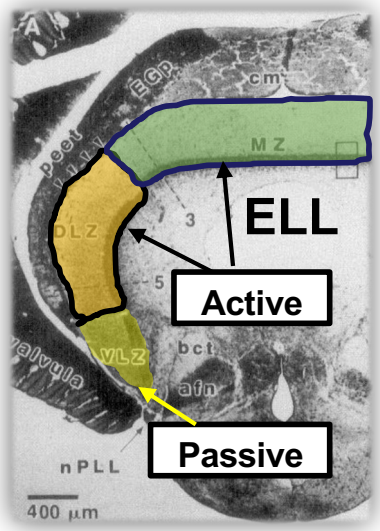


- Passive and Active

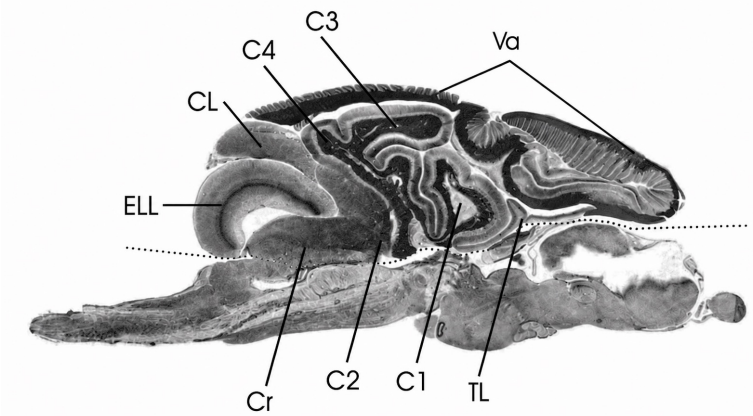


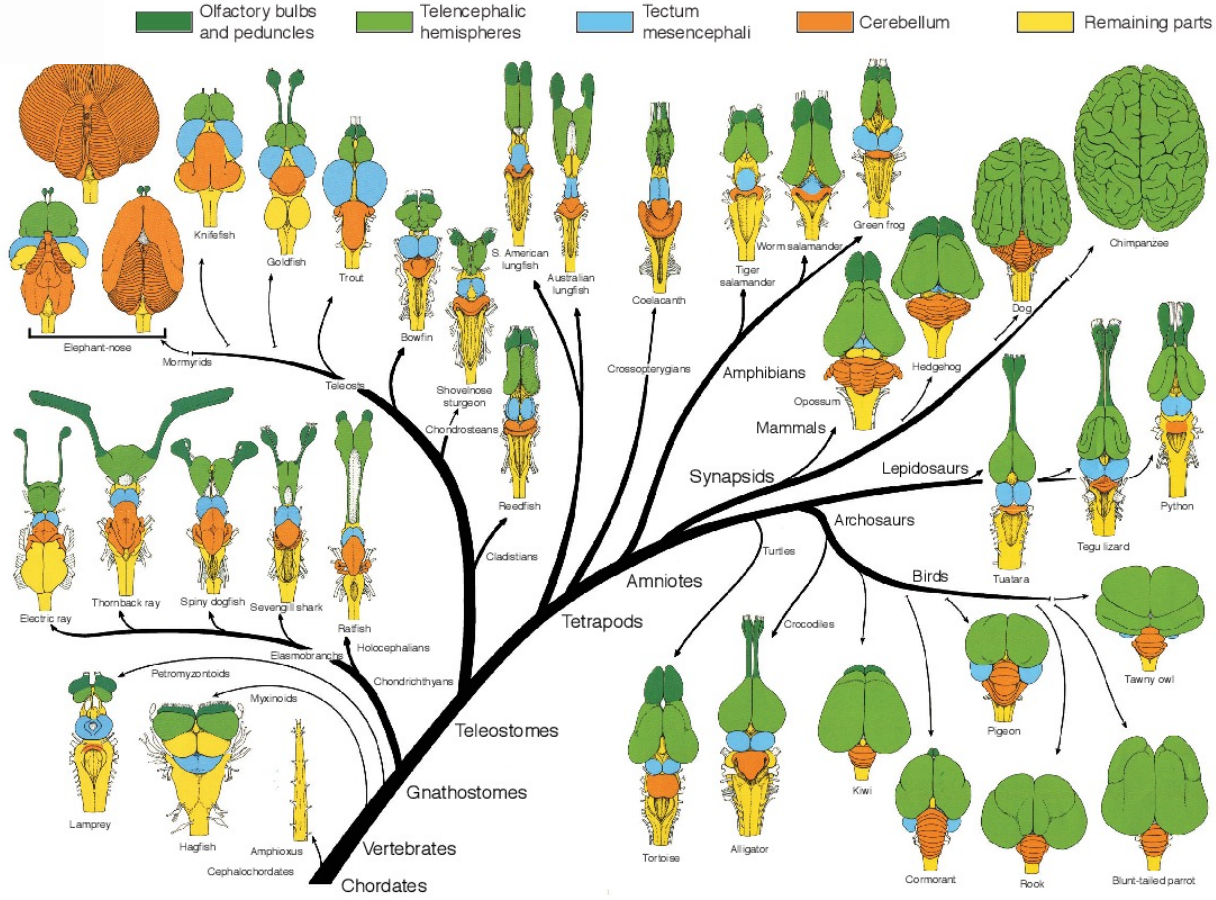
passive
low-frequency

active
tuned to EOD

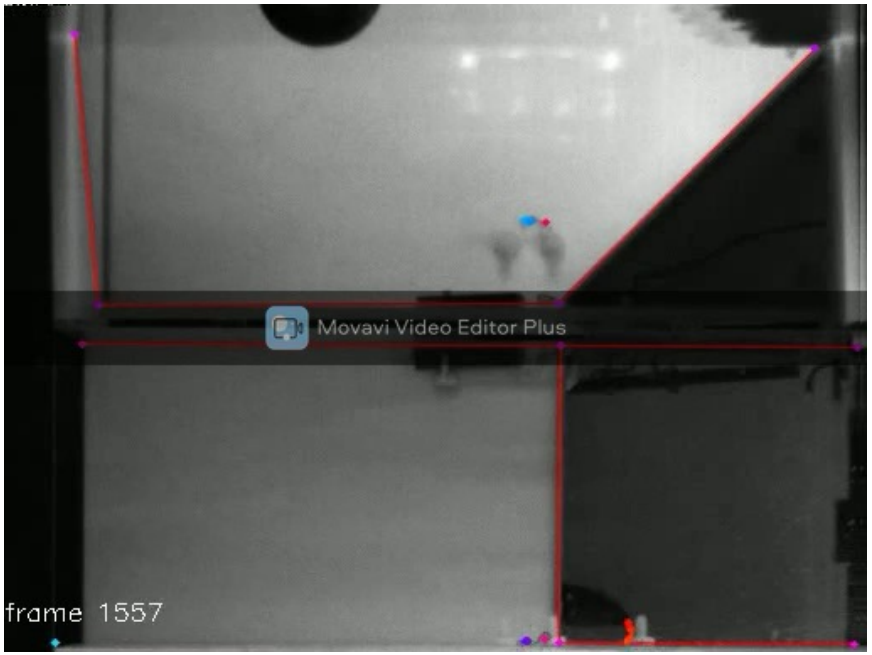


Cerebellar Champion





Electrolocation (pulse type fish)



Schnauzenorgan



Elephantnose fish
(*Gnathonemus petersii*)
pulse type, Africa

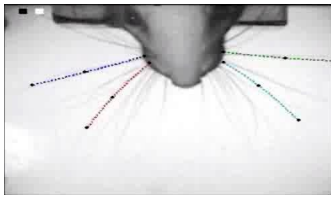
Active Sensing



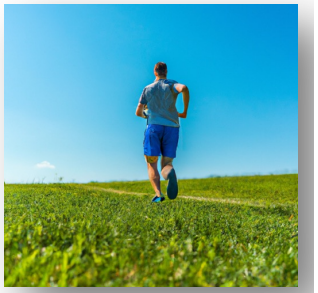
David Kleinfeld

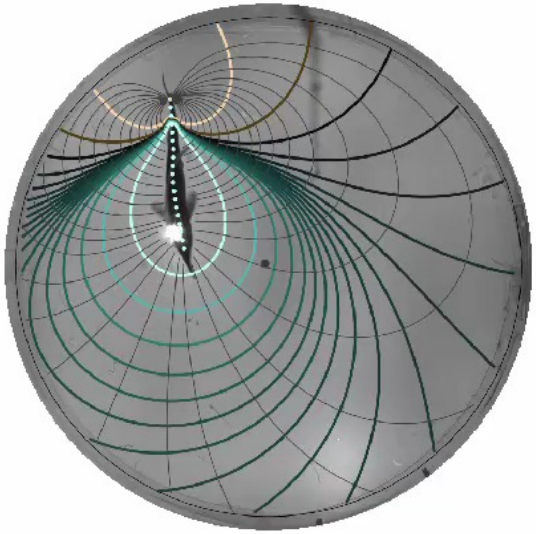


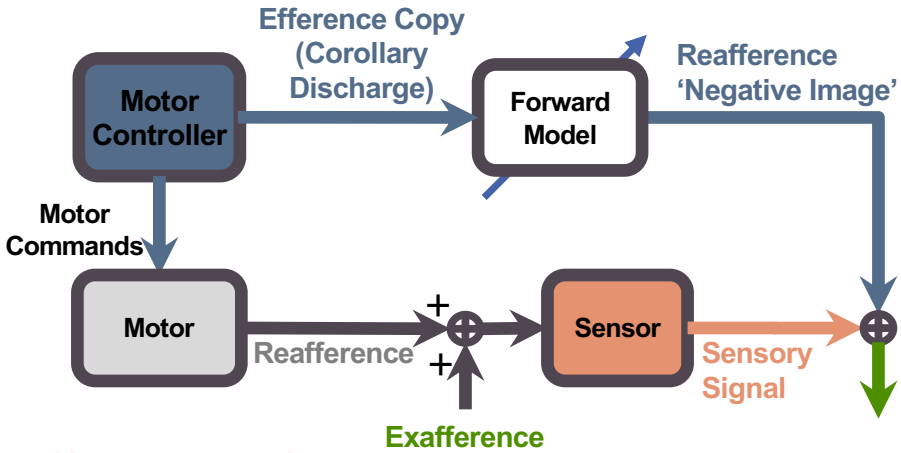




Ehud Ahissar





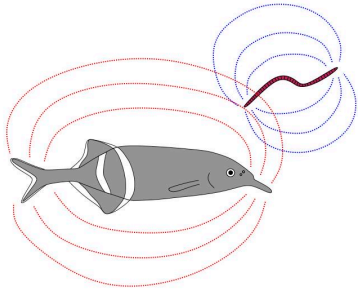


*"We shall propose that the **efference** leaves an "**image**" of itself somewhere in the CNS, to which the reafference… [is] the **negative**… so that when superimposed the image disappears."*
Von Holst, 1954

Reafference: self-generated sensations
Exafference: externally generated sensations

Cancellation of Self-Generated Sensations

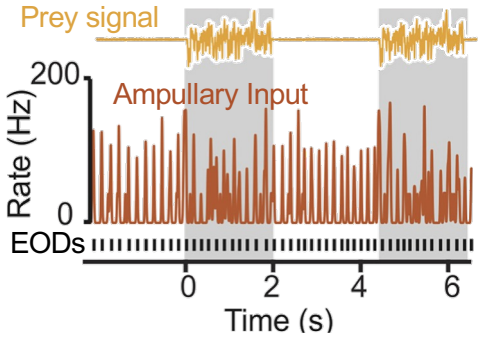
Passive electrolocation

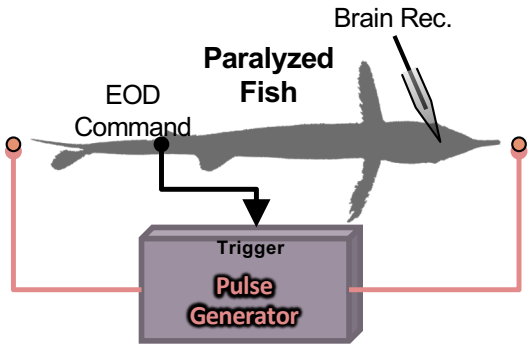


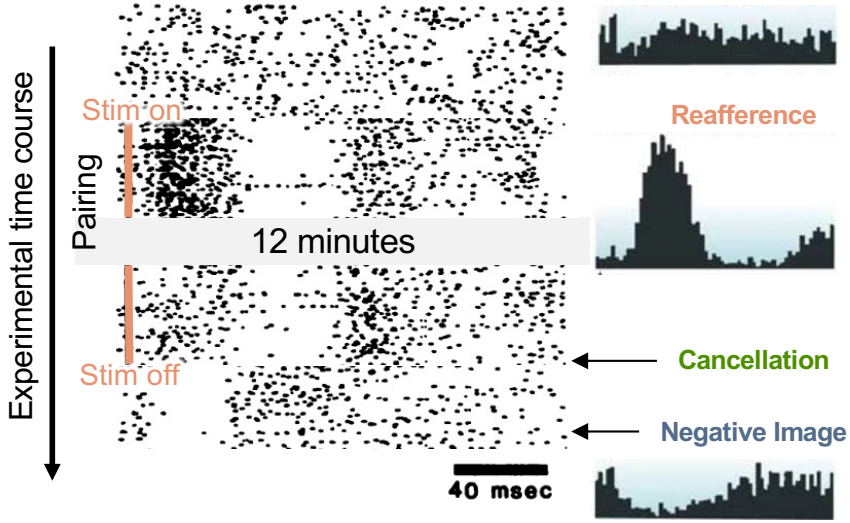
Signal = Electric field of aquatic animals

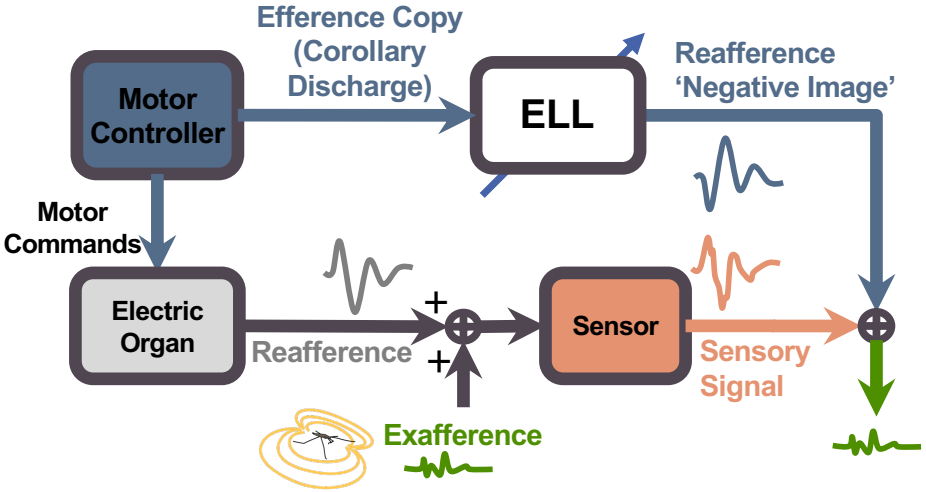
Noise = Own EOD

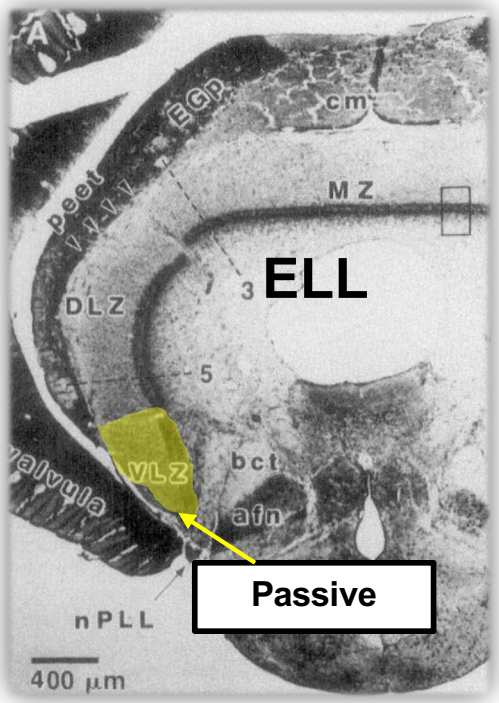
Enikolopov, 2018, *Neuron*



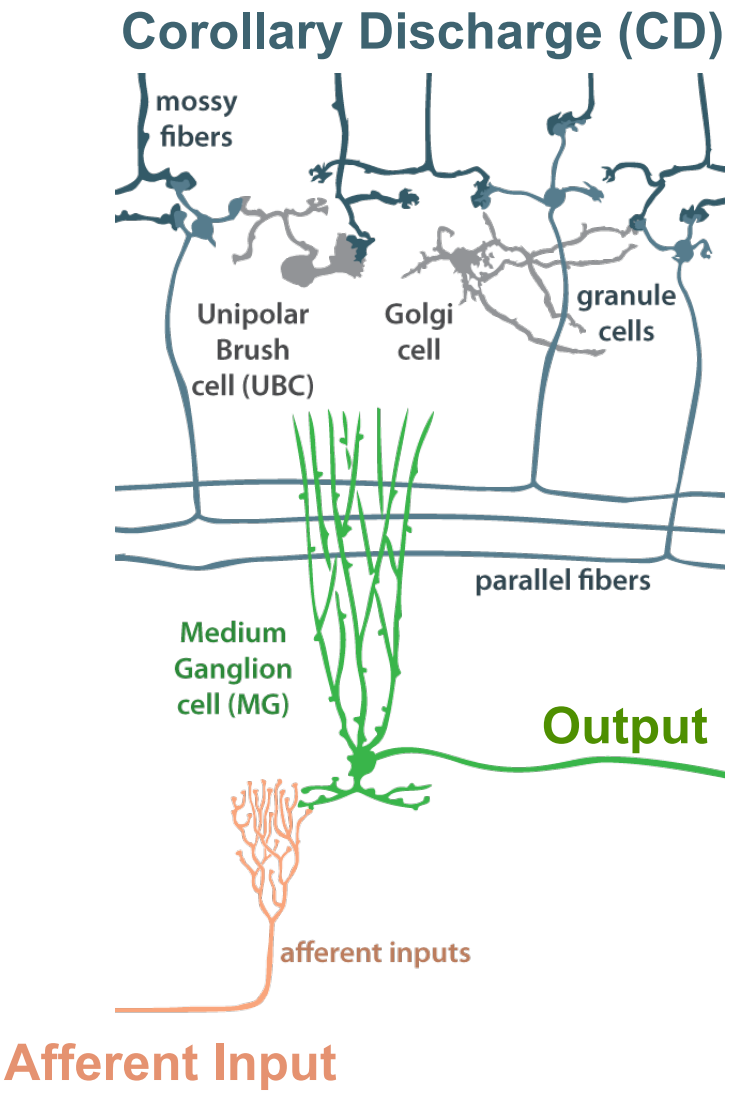


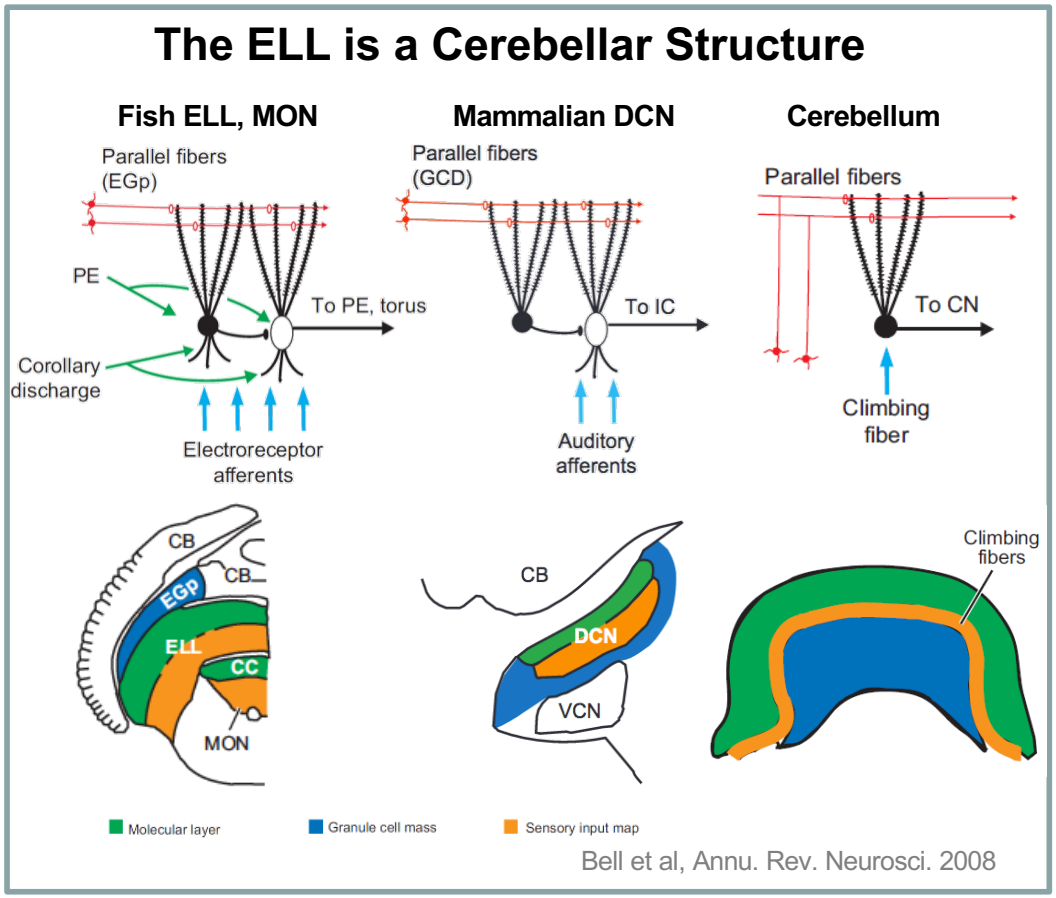


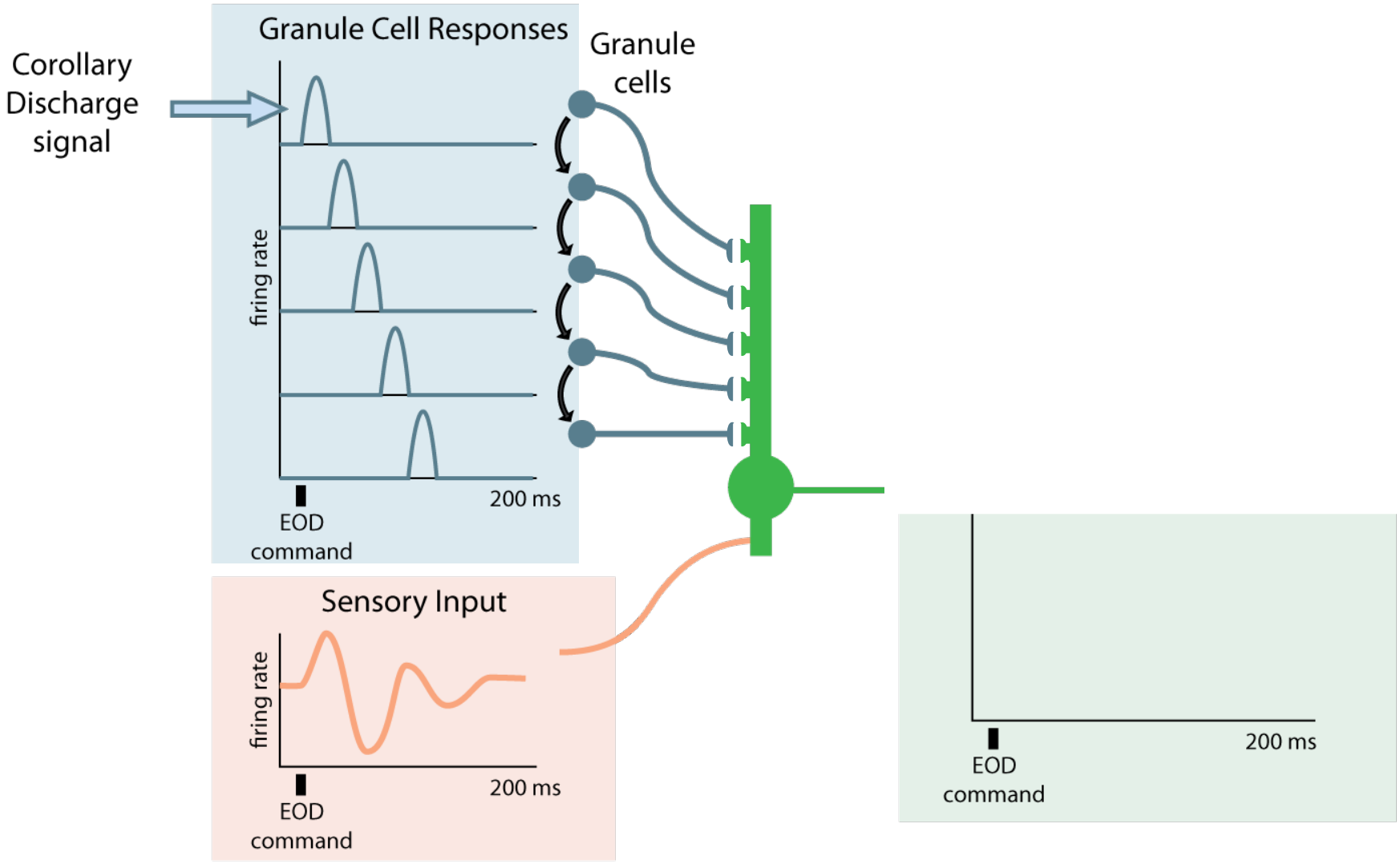




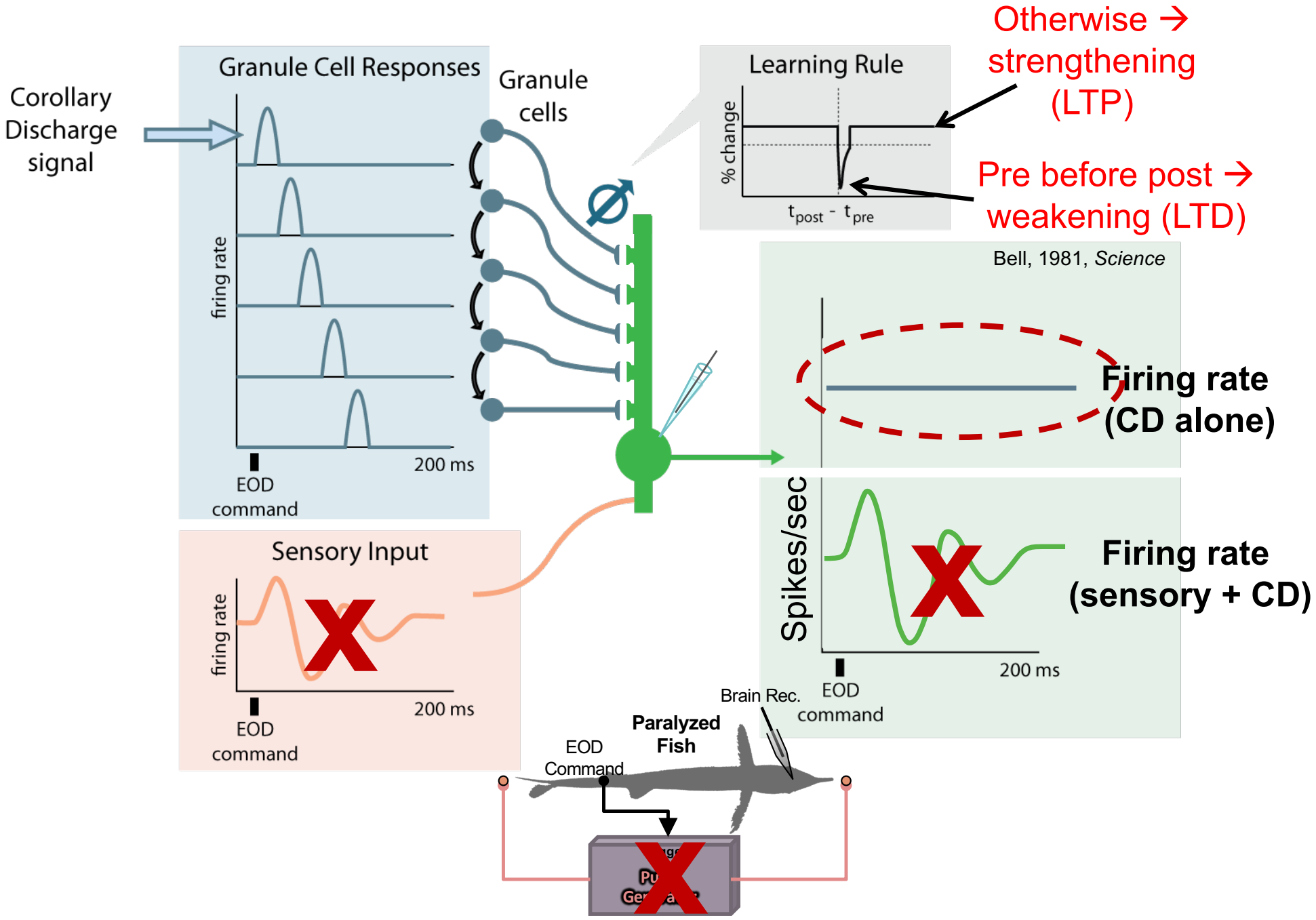
The ELL: Adaptive Reafference Filter

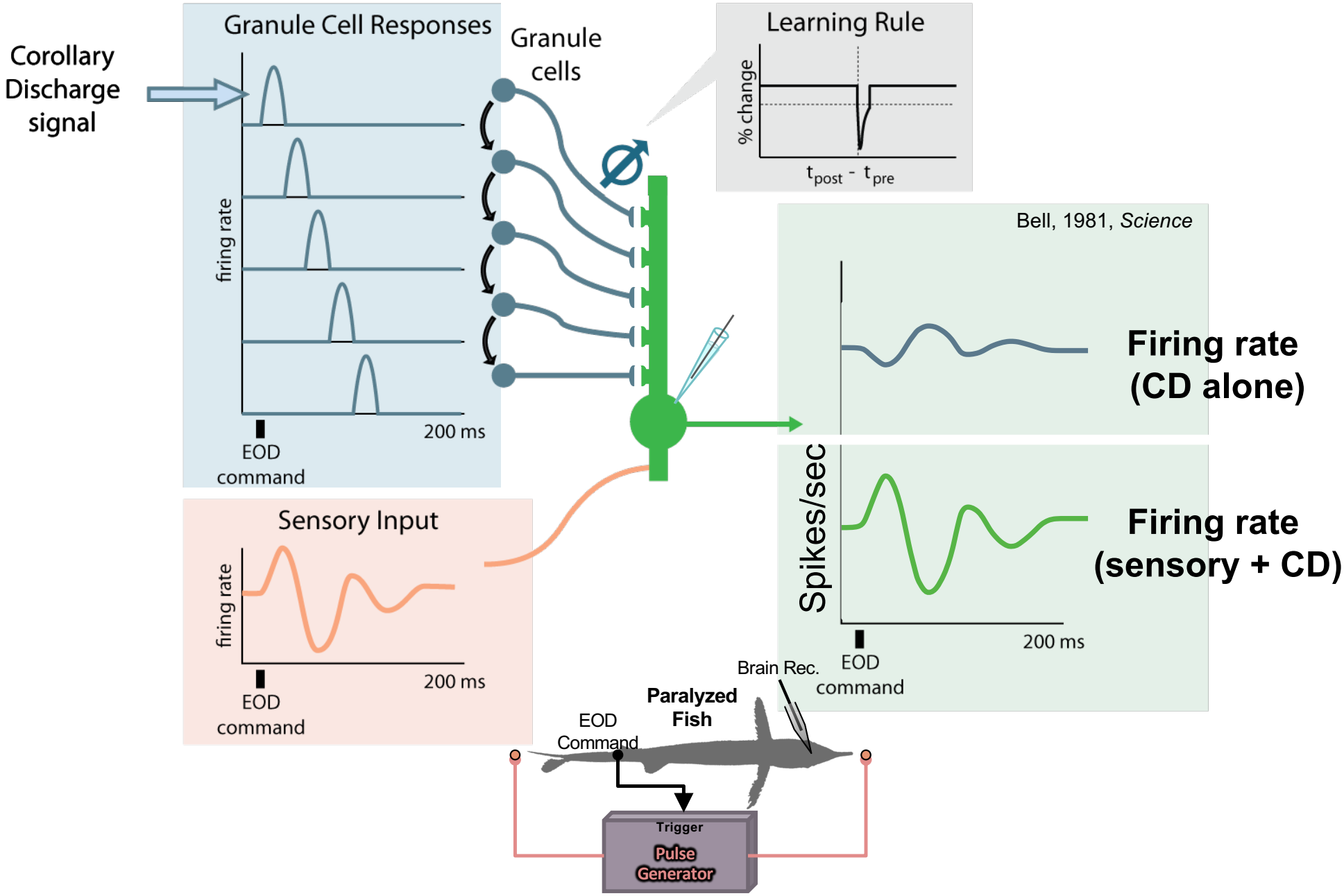


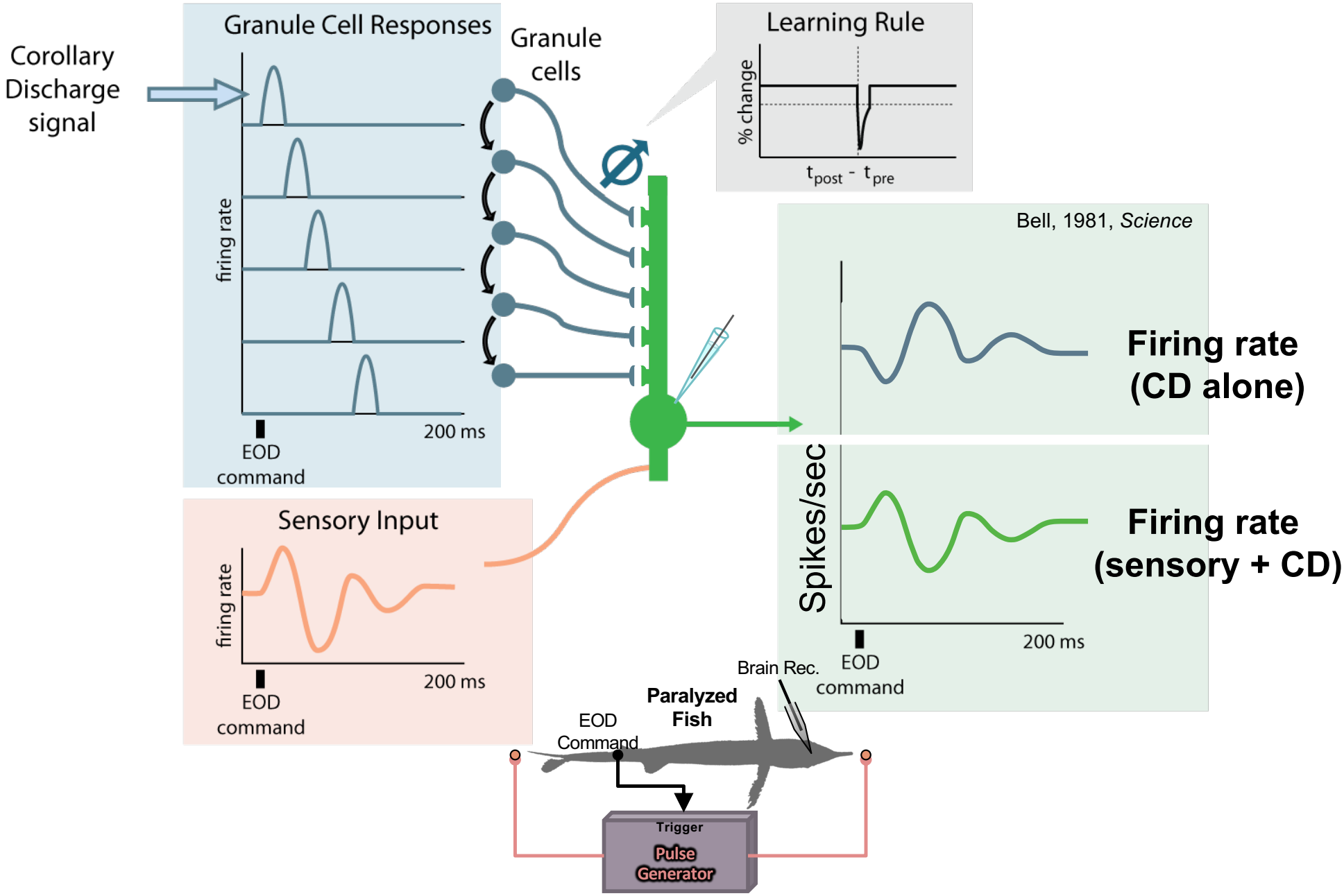


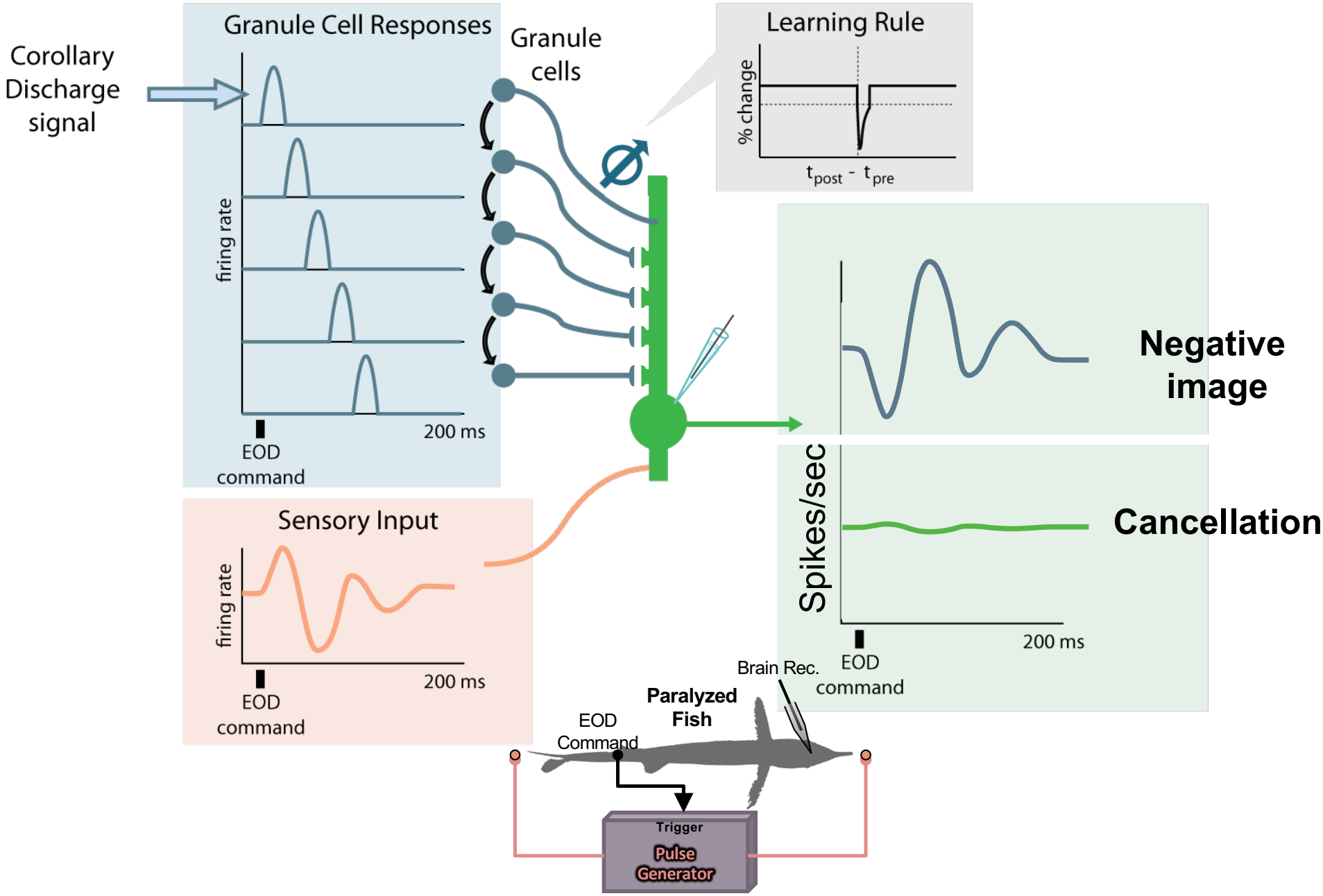

Corollary Discharge signal
Granule Cell Responses
Granule cells
firing rate
200 ms
EOD command
Sensory Input
firing rate
200 ms
EOD command
200 ms
EOD command

Anti-Hebbian Spike-Timing Dependent Plasticity



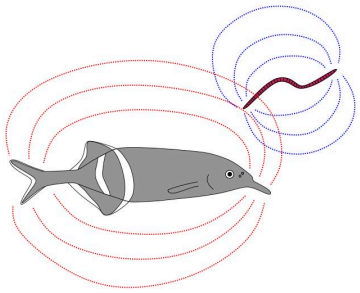

Corollary Discharge signal
Granule Cell Responses
firing rate
200 ms
EOD command
Granule cells
Learning Rule
% change
t_{post} - t_{pre}
Bell, 1981, *Science*
Firing rate (CD alone)
Spikes/sec
200 ms
EOD command
Firing rate (sensory + CD)
Sensory Input
firing rate
200 ms
EOD command
Brain Rec.
Paralyzed Fish
EOD Command
Trigger
Pulse Generator


Corollary Discharge signal
Granule Cell Responses
Granule cells
firing rate
200 ms
EOD command
Learning Rule
% change
t_{post} - t_{pre}
Bell, 1981, *Science*
Firing rate (CD alone)
Spikes/sec
200 ms
EOD command
Firing rate (sensory + CD)
Sensory Input
firing rate
200 ms
EOD command
Brain Rec.
Paralyzed Fish
EOD Command
Trigger
Pulse Generator


Corollary Discharge signal
Granule Cell Responses
firing rate
200 ms
EOD command
Granule cells
Learning Rule
% change
t_{post} - t_{pre}
Negative image
Spikes/sec
200 ms
EOD command
Cancellation
Sensory Input
firing rate
200 ms
EOD command
Brain Rec.
Paralyzed Fish
EOD Command
Trigger
Pulse Generator

Reafference Cancellation in Active Electrolocation

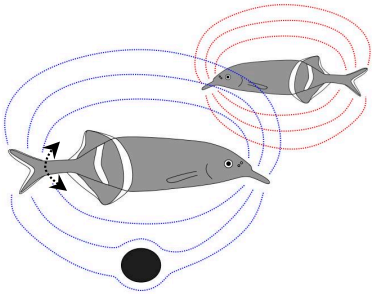
Passive electrolocation



Signal = Electric field of aquatic animals

Noise = Own EOD

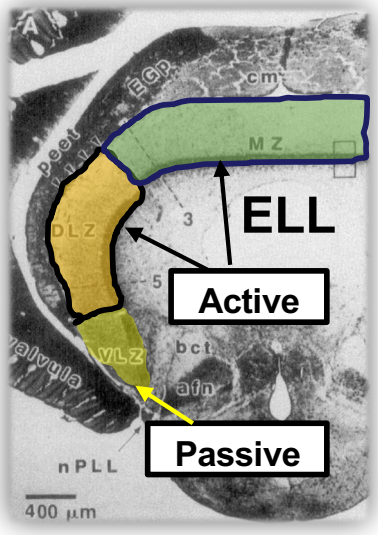
Active electrolocation

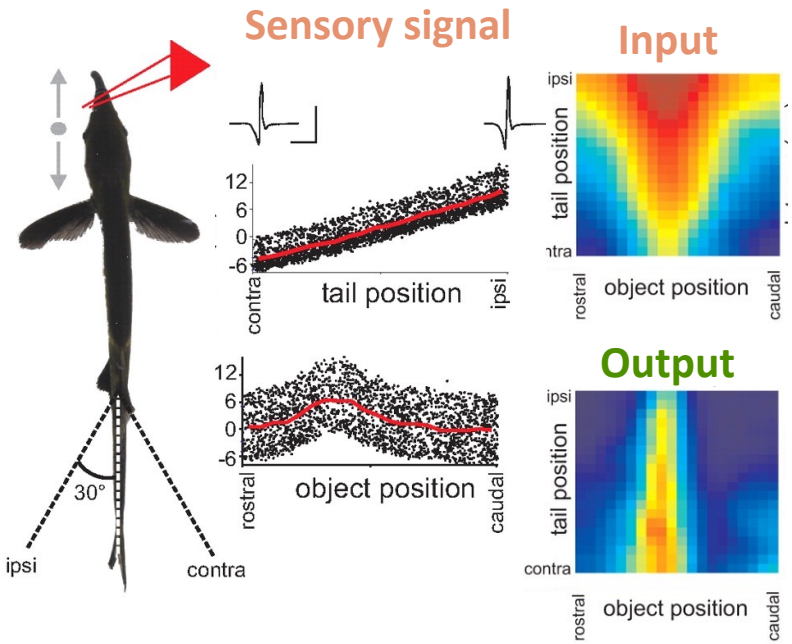


Signal = Own EOD

Noise = Other fish's EOD

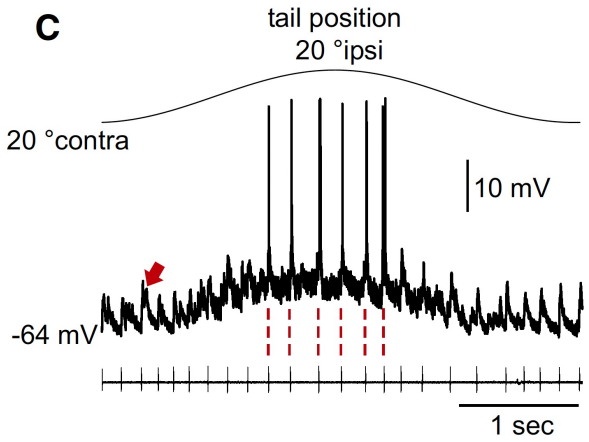
Self-motion



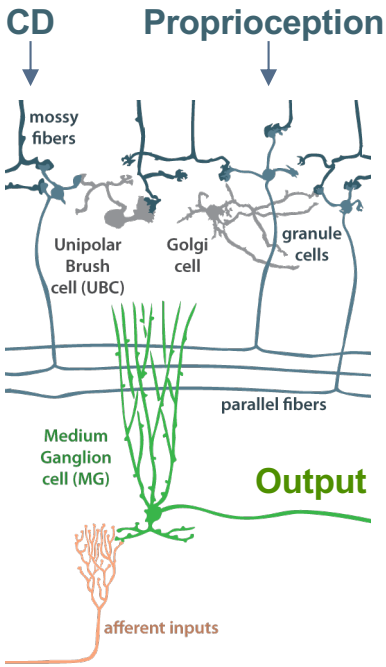


Sawtell & Williams, 2008, *JNS*

GC combining tail proprioception + CD

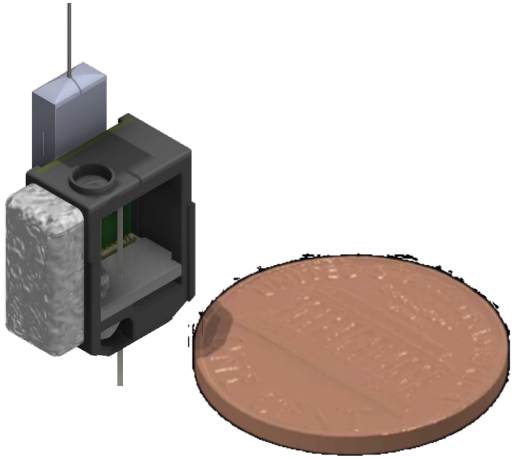


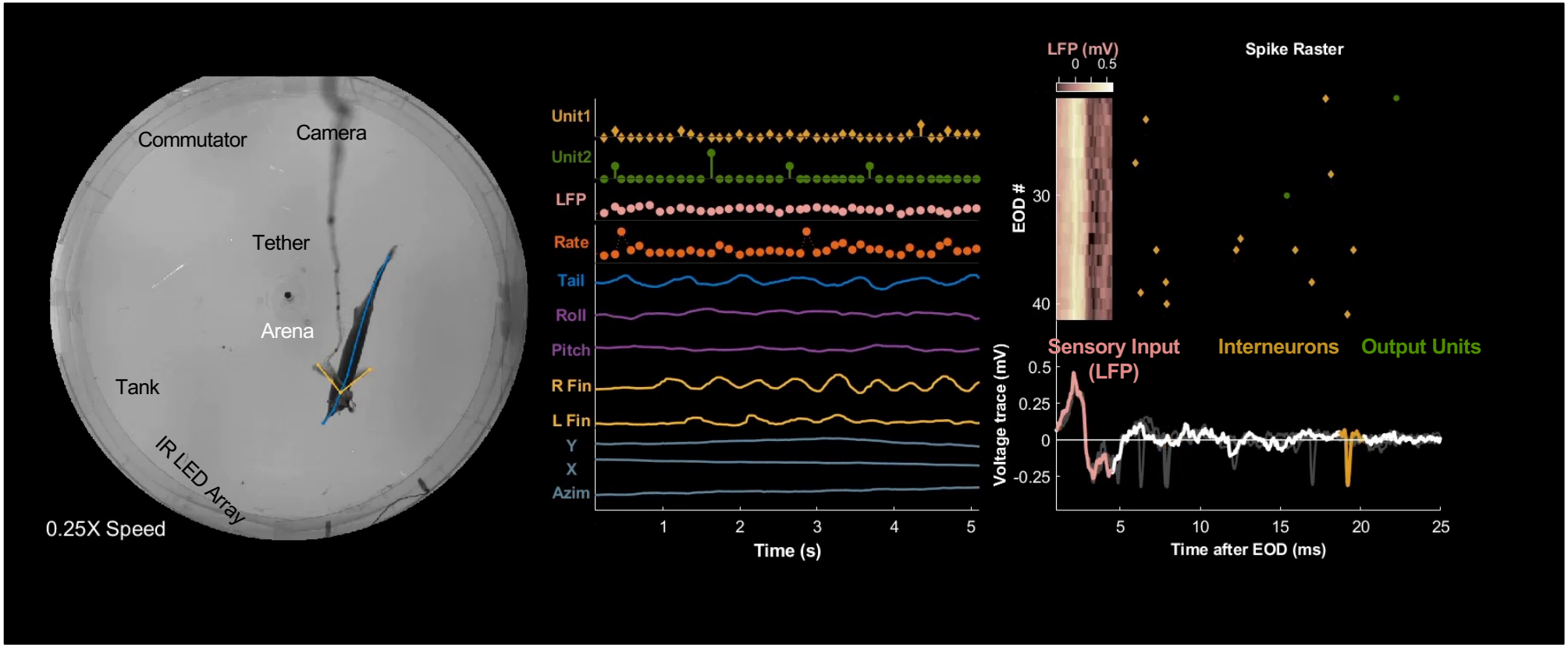
Sawtell, 2010, *Neuron*



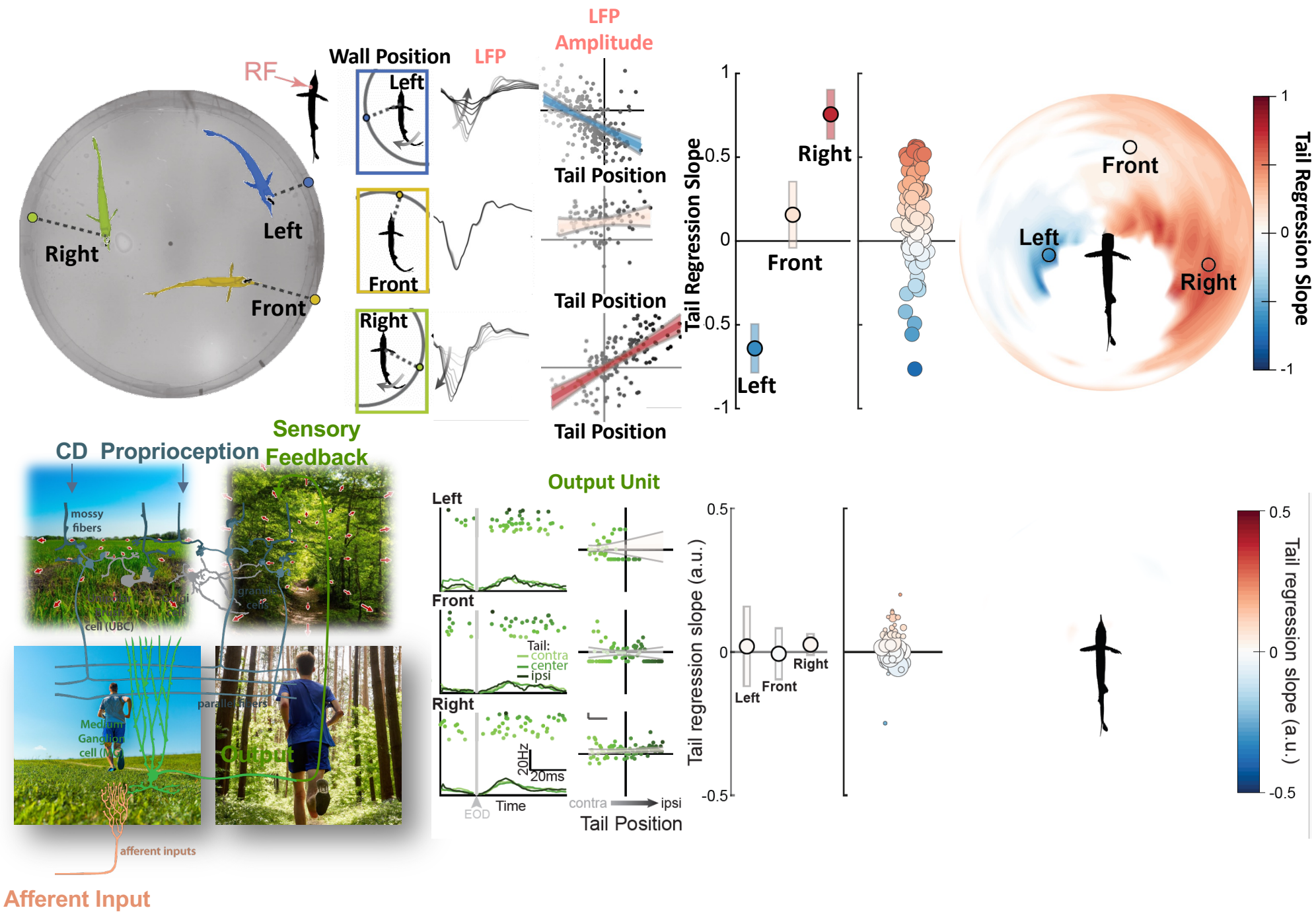
Afferent Input

Electrolocation in Freely Swimming Fish



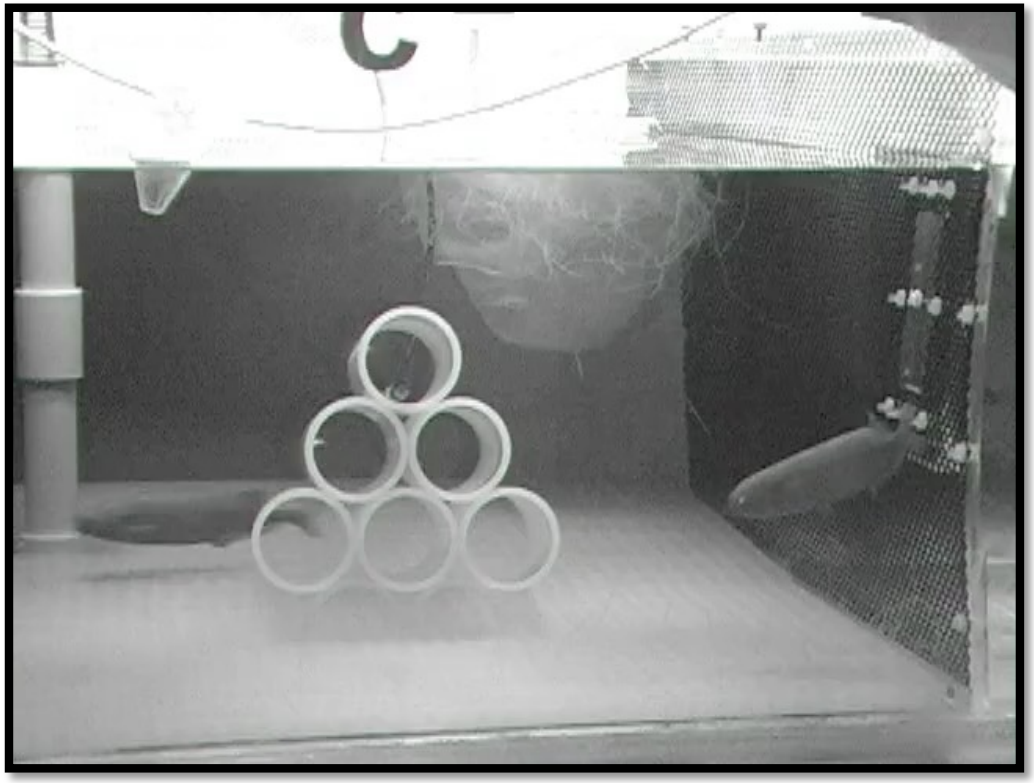


Wallach & Sawtell, *Neuron*, 2023


RF
Wall Position
LFP
LFP Amplitude
Left
Front
Right
Tail Position
Tail Regression Slope
Sensory Feedback
CD
Proprioception
mossy fibers
granule cells
cell (UBC)
parallel fibers
Medium Ganglion cell (MG)
Output
afferent inputs
Afferent Input
Output Unit
Tail: contra center ipsi
20Hz
20ms
EOD
Time
contra
ipsi
Tail regression slope (a.u.)

Electrocommunication

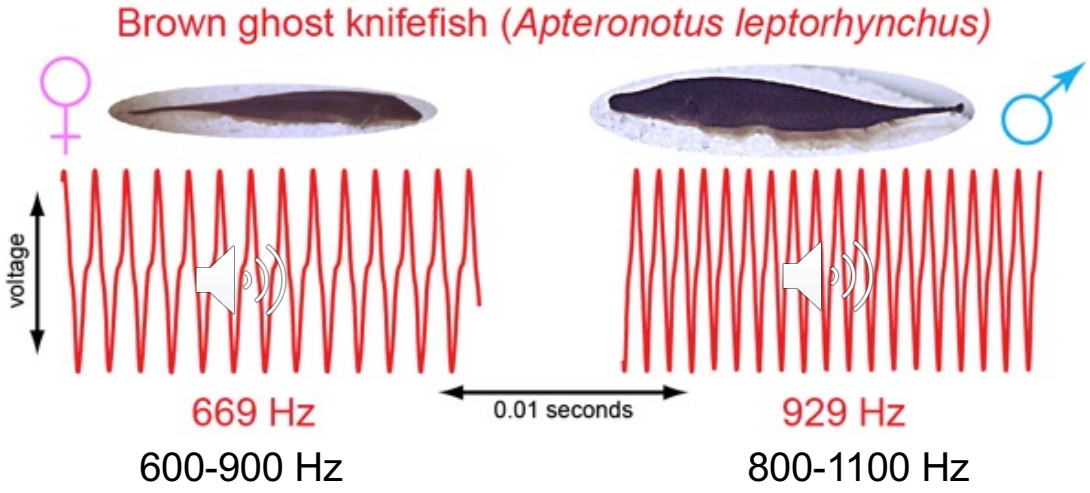
Electric fish use their discharge to communicate courtship, aggression, dominance etc

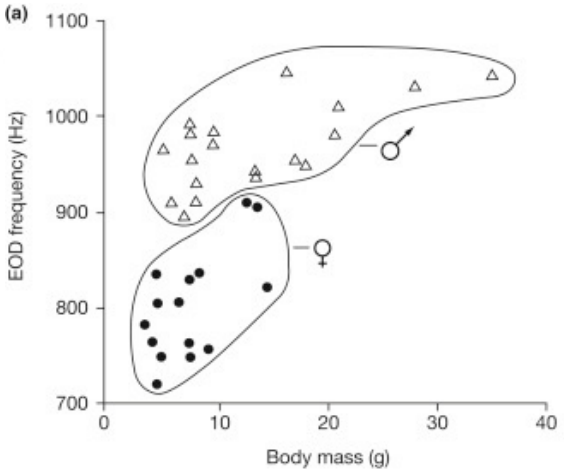


Very complicated in pulse fish

Electrocommunication (wave type fish)

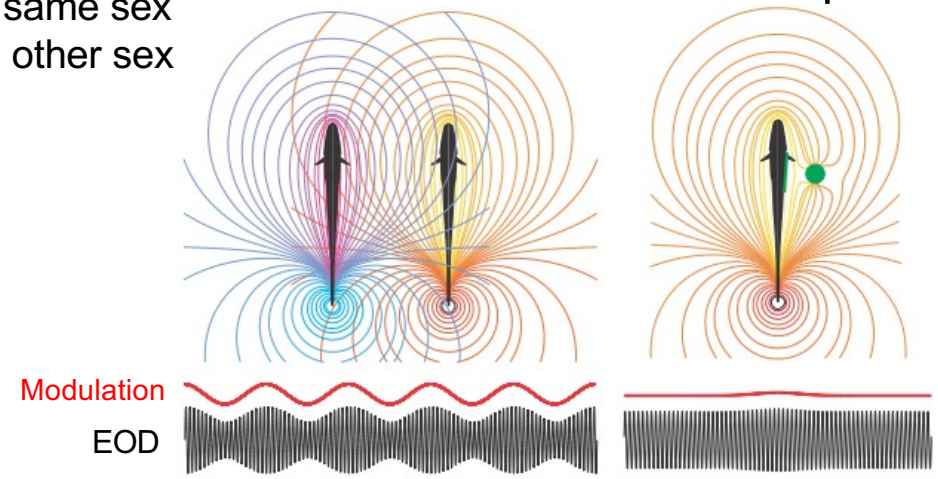
EOD frequency conveys sex, status



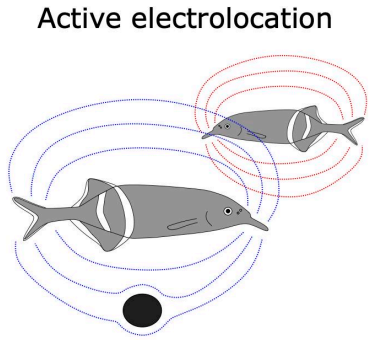


Beat: interference pattern between two EODs

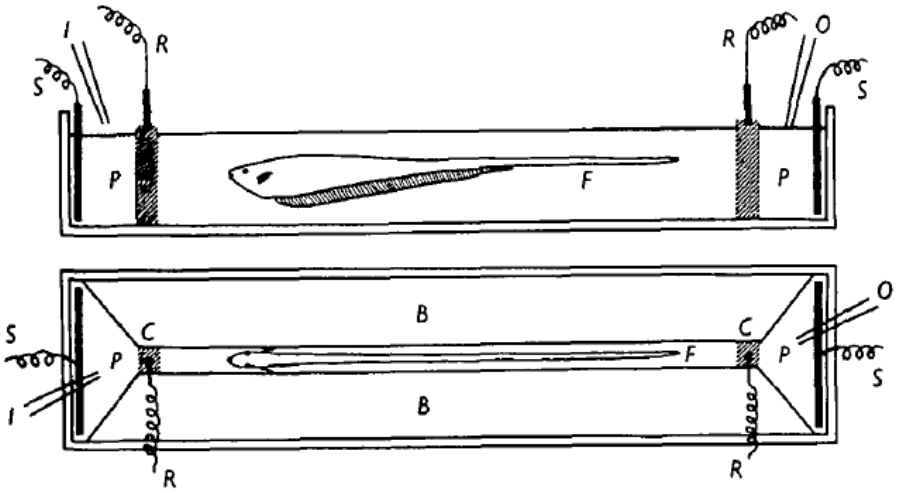
- Beat frequency = EOD freq. difference
  - Low beat freq = same sex
  - High beat freq = other sex

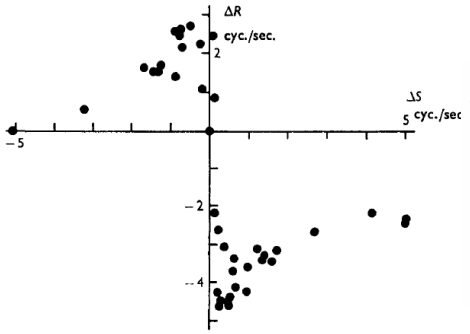


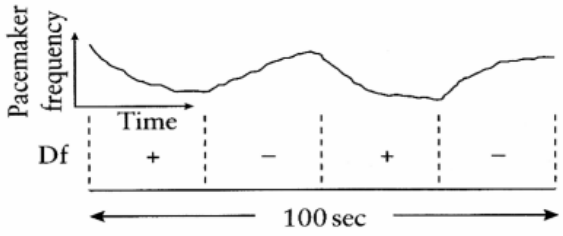
Jamming:
Low-frequency beat interferes with perception of objects



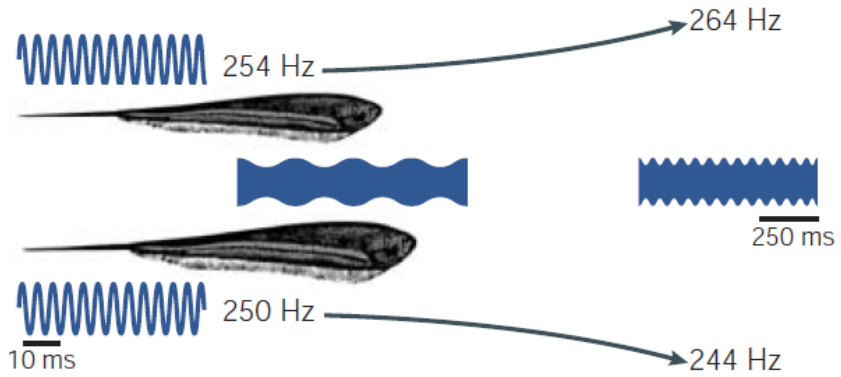
Jamming avoidance response

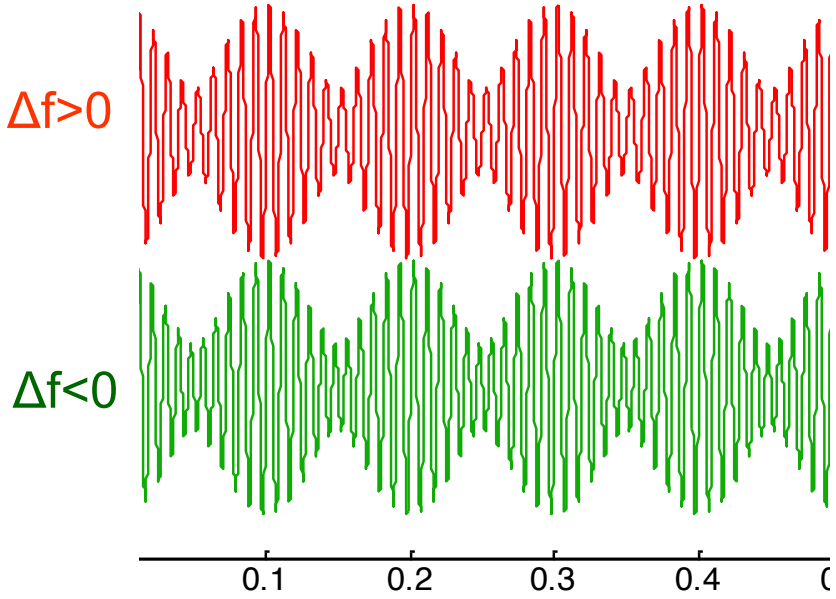






Watanabe & Takeda, J. Exp. Biol. 1963







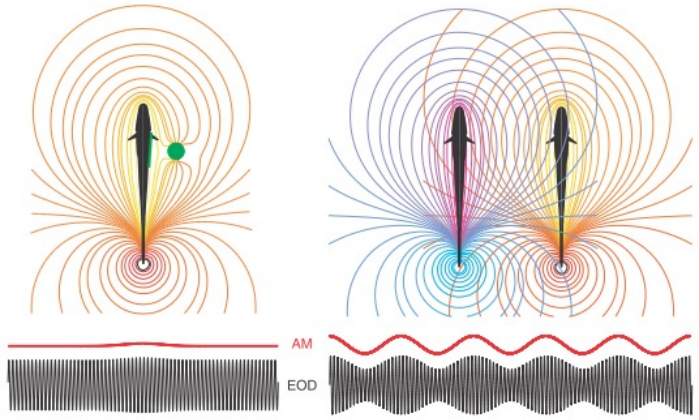
- Reproducible sensory-motor behavior of ethological importance.
- Requires non-trivial neuronal computation

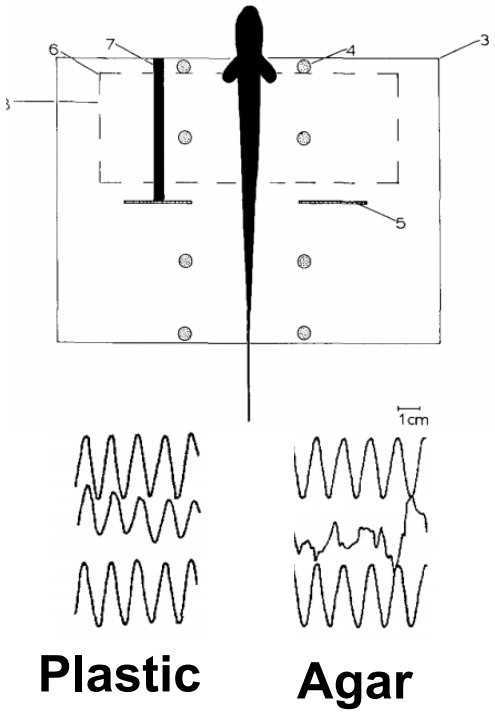
Jamming avoidance response

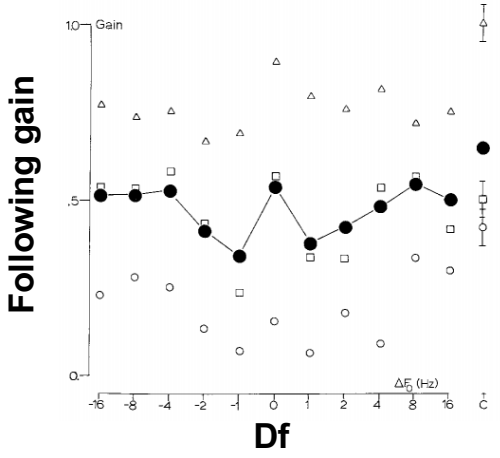
- Presence of external objects exerts low frequency (<20Hz) modulations of the EOD ‘image’ on the skin.

- Neighboring fish with similar EODr cause low-frequency interference which ‘jams’ perception.

- Solution: each fish alters its EOD rate so that the frequency difference will be outside the perceptual band (>20Hz)

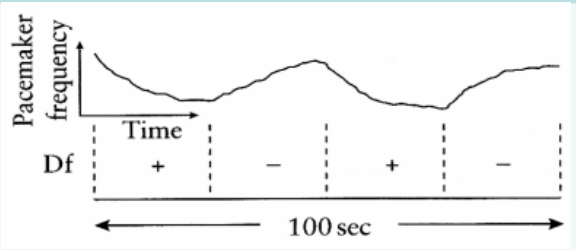
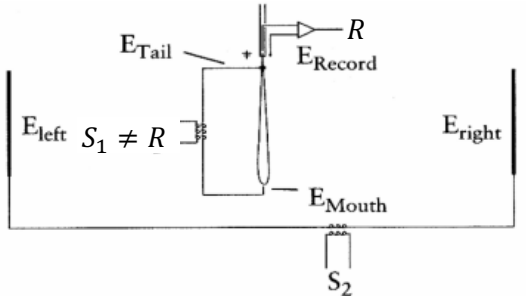
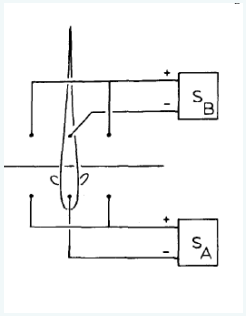
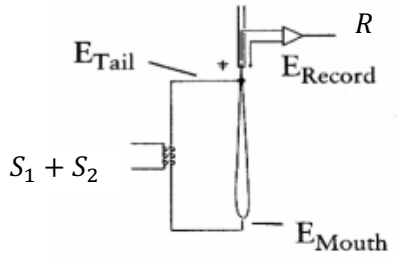






Heiligenberg J Comp Physiol. 1973,
Matsubara & Heiligenberg J Comp Physiol 1978

Possible mechanisms

Mechanism	Evidence	Experiment
Iterative search	✗ Immediate correct response	
Efference copy ✗ Open loop response according to mimic frequency ✗ No PPn→ELL pathway		
Using temporal interference pattern	✓ JAR abolished in split chamber stimulation	
Using spatial interference pattern	✓ Incorrect JAR in 'identical geometry' stimulation	

JAR computation

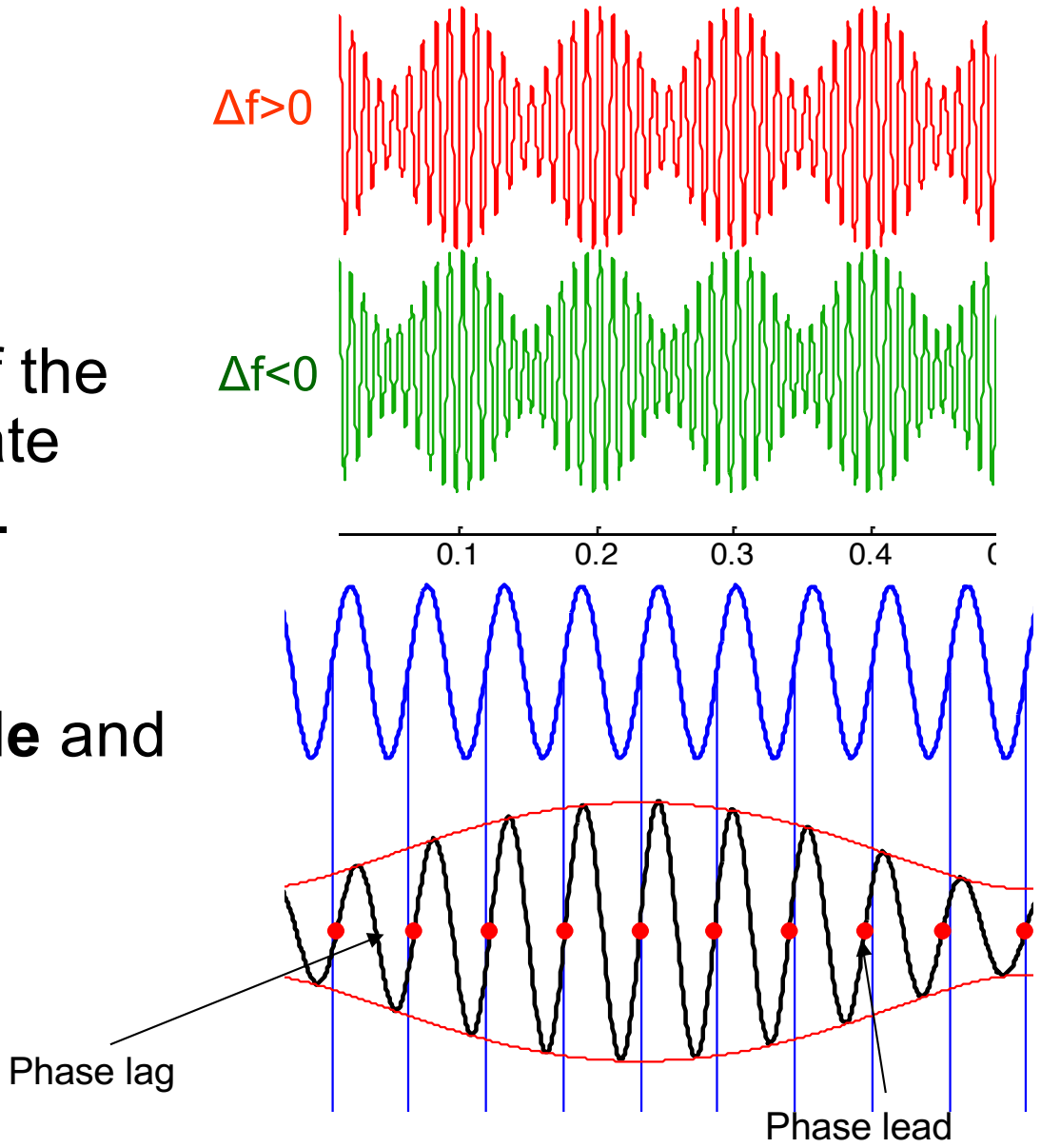
- f_1- Fish's own EOD rate
  f_2- Interefence EOD rate

  $\Delta f = f_2 - f_1$

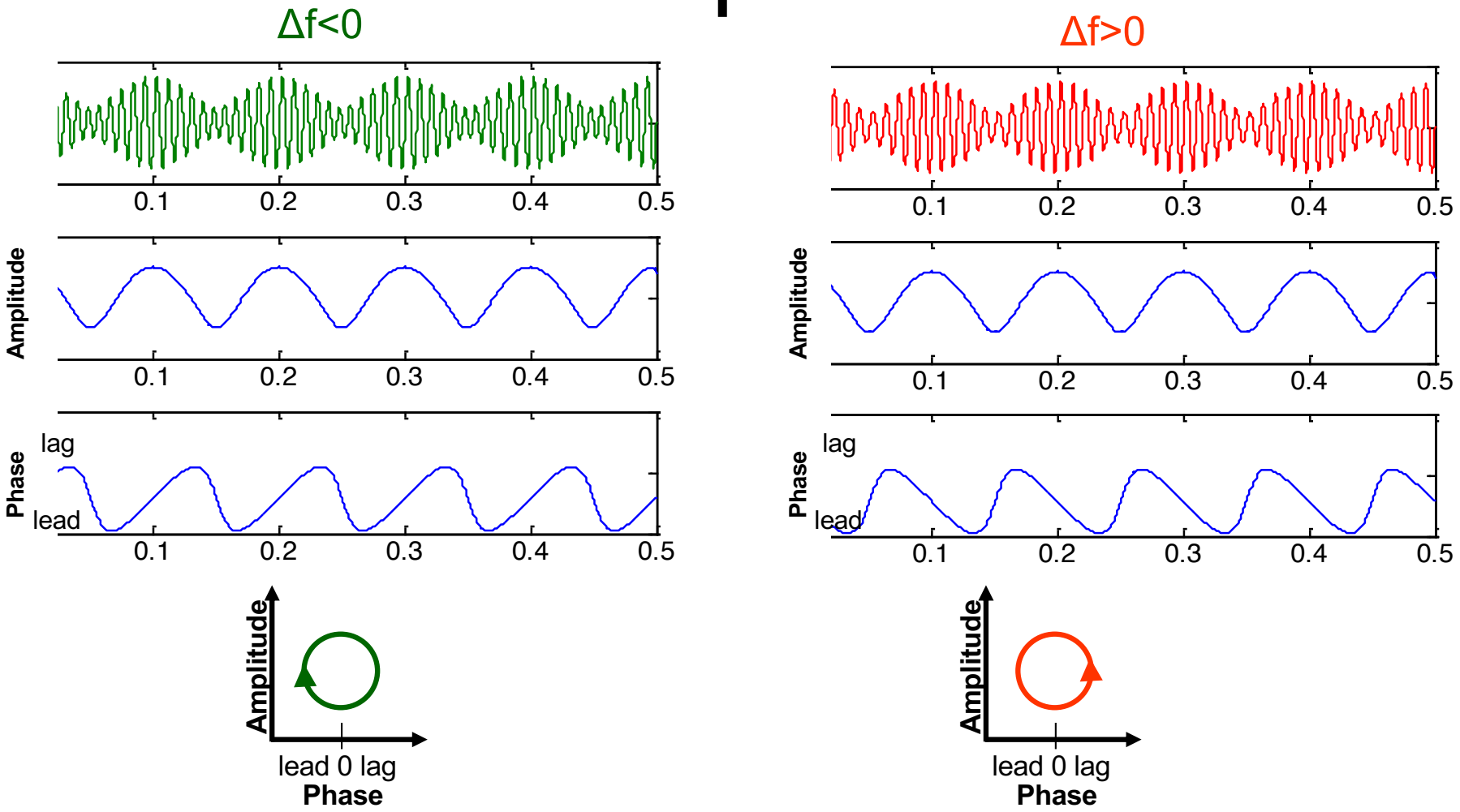
- The fish needs to compute if the neighbor is using a higher rate ($\Delta f>0$) or a lower rate ($\Delta f<0$).

- The interference causes a modulation in both **amplitude** and **phase** called **'beat'.**

- The rate of the beat is $|\Delta f|$

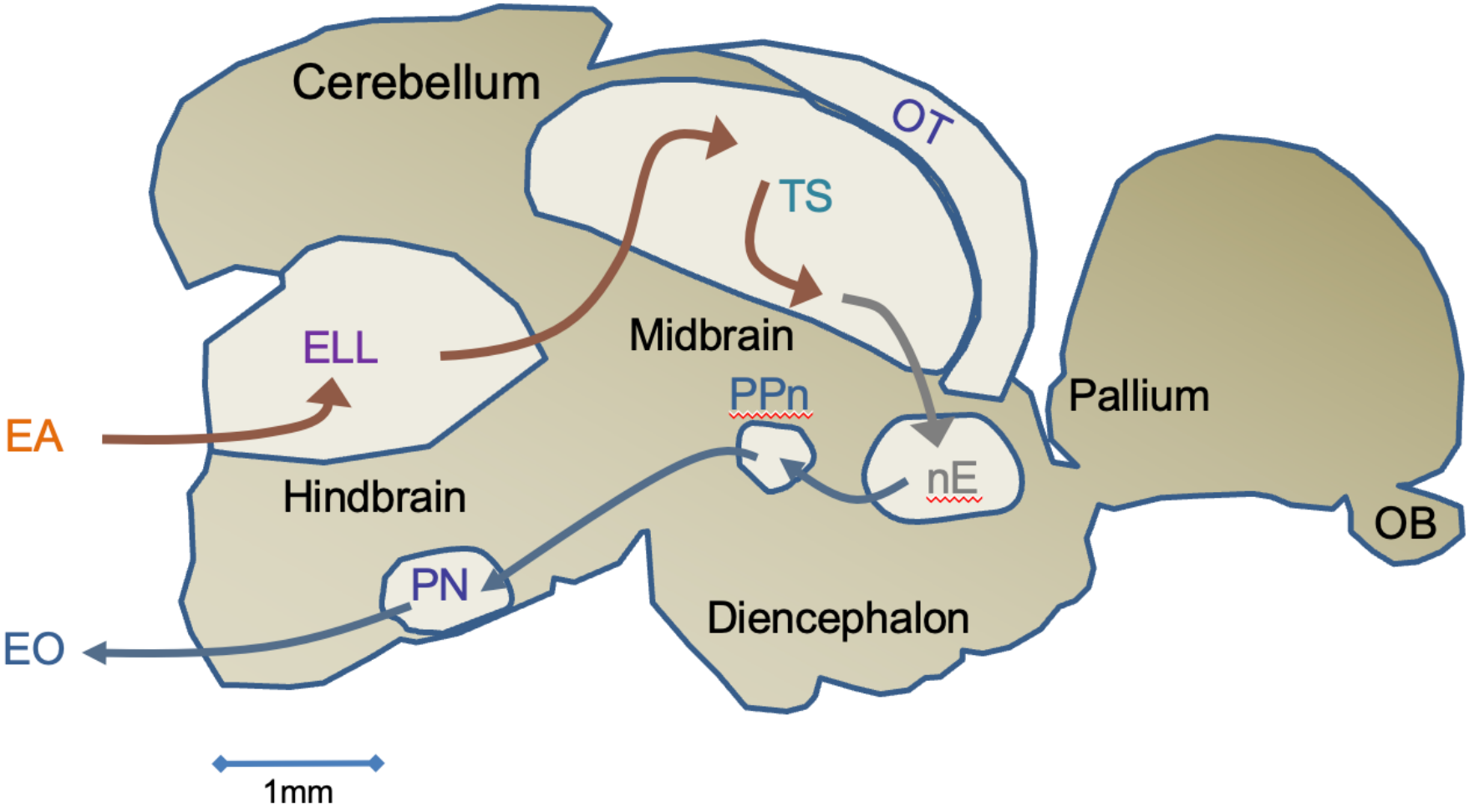


JAR computation

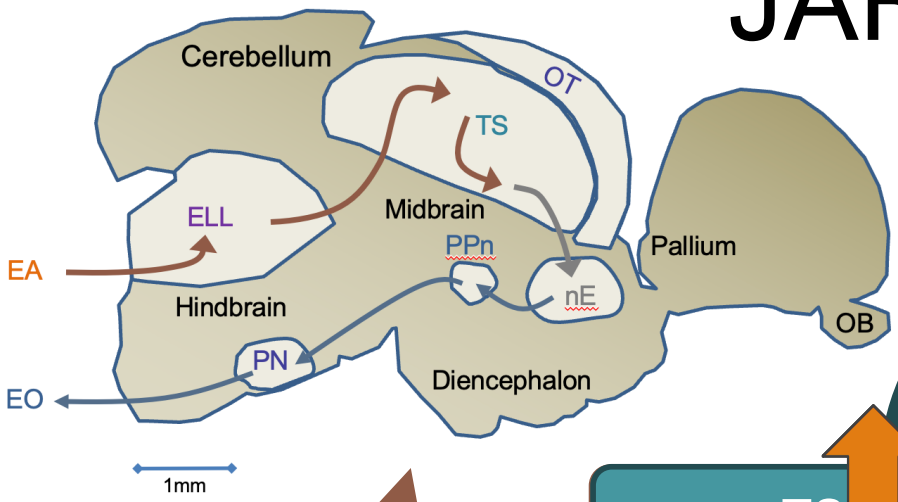


- Pattern of modulation in phase and amplitude holds information about sign of Δf:
  - When Δf>0 phase lag during amplitude rise and vice-versa.
  - When Δf<0 phase lead during amplitude rise and vice-versa.
- Rotations in different directions in amplitude/phase state-plane.

JAR circuit


Cerebellum
OT
TS
ELL
Midbrain
PPn
Pallium
EA
nE
Hindbrain
OB
PN
Diencephalon
EO
1mm

JAR circuit



n. Electrosensorius

Δf<0

nE Up

nE Down

Δf>0

TS 3b,c

SPPn

PPnG

Pn Pacemakers

Pn Relay

Electric Organ

Sensory

Midbrain (Torus)

Medulla (ELL)

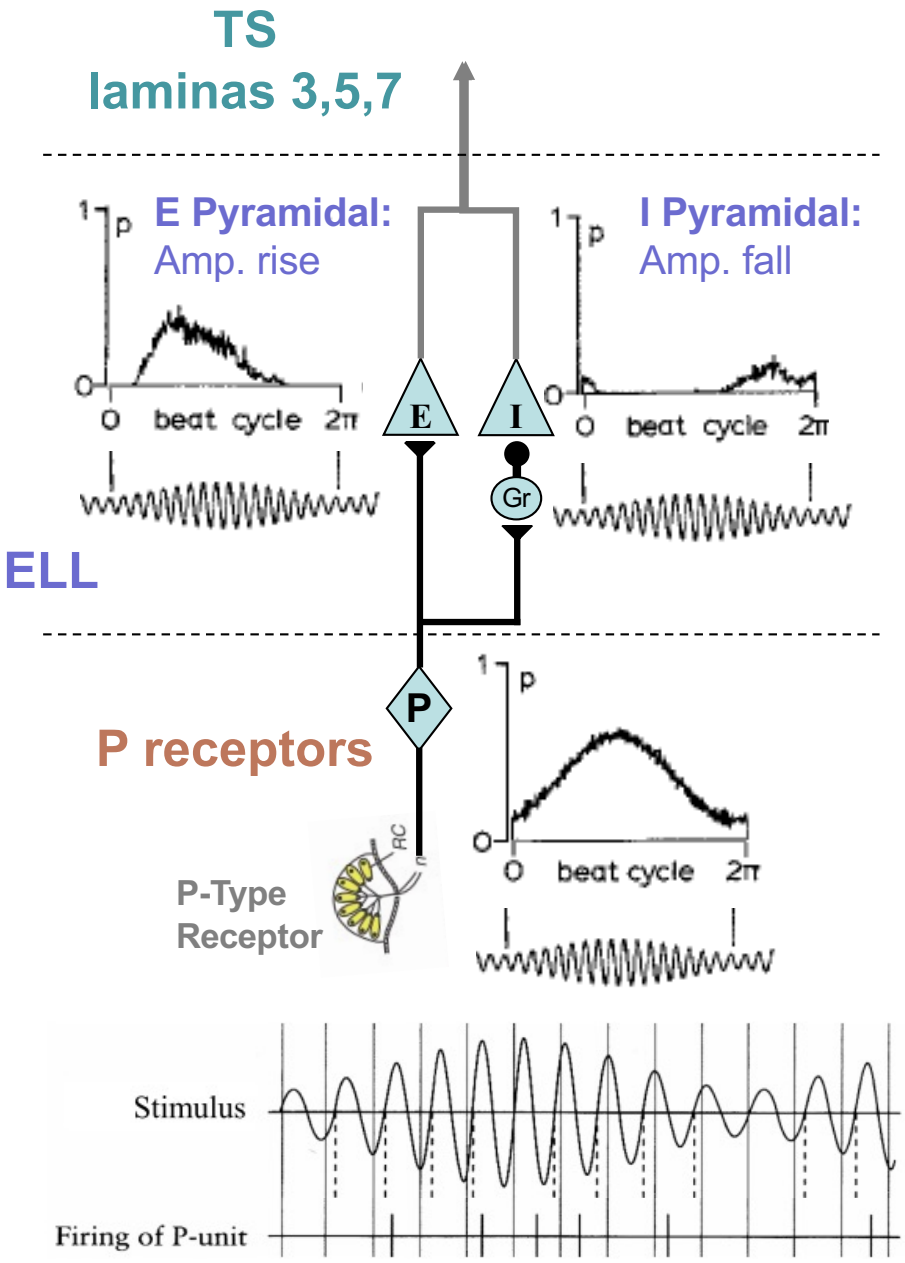
Amplitude

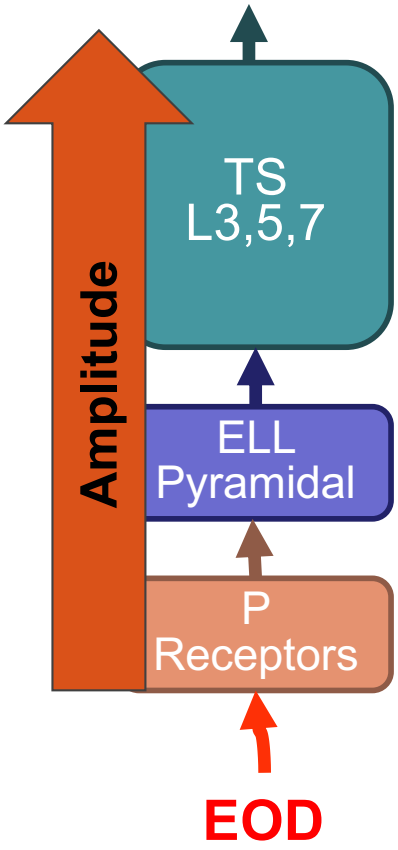
Phase

Motor

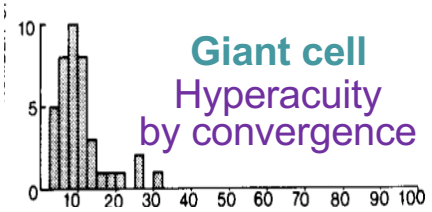
EOD

JAR physiology: Amplitude coding





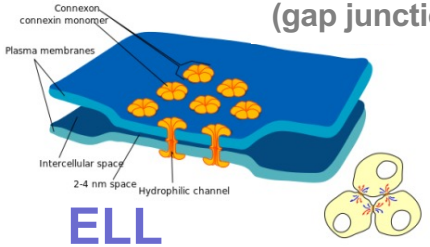
JAR physiology: Phase coding



Giant cell
Hyperacuity
by convergence

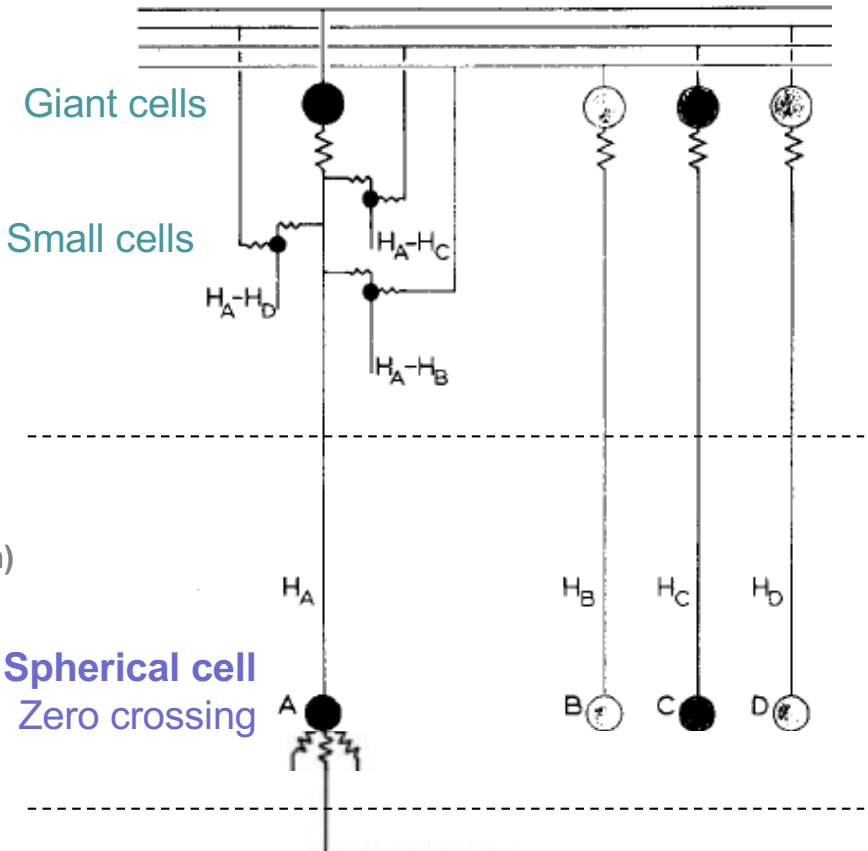
S.D. Spike Latency (usec)

TS
lamina 6

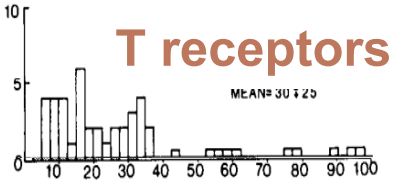


ELL

Electrical
Synapse
(gap junction)



Spherical cell
Zero crossing

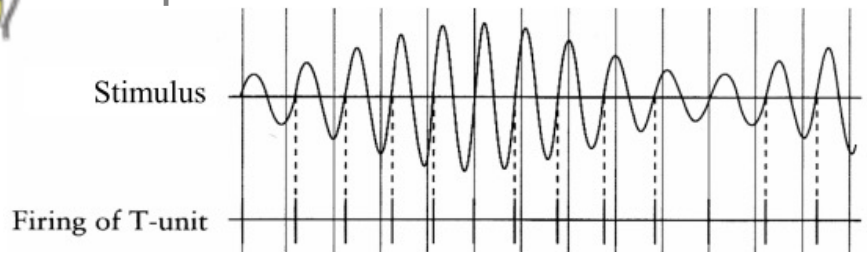


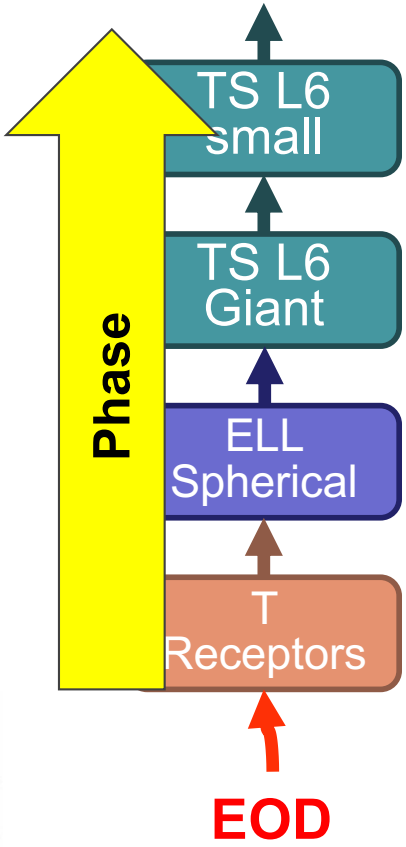
T receptors

S.D. Spike Latency (usec)



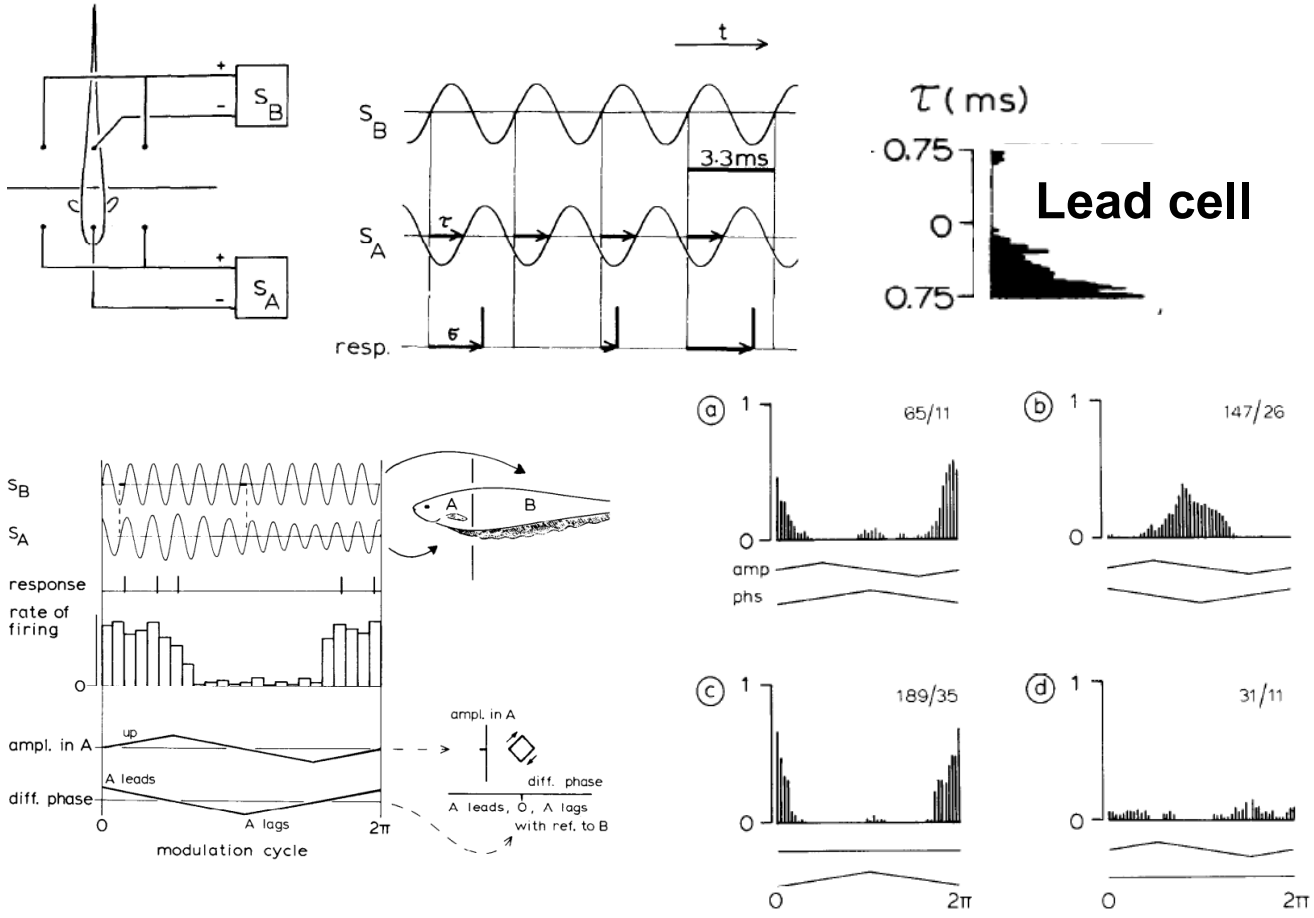
T-Type
Receptor

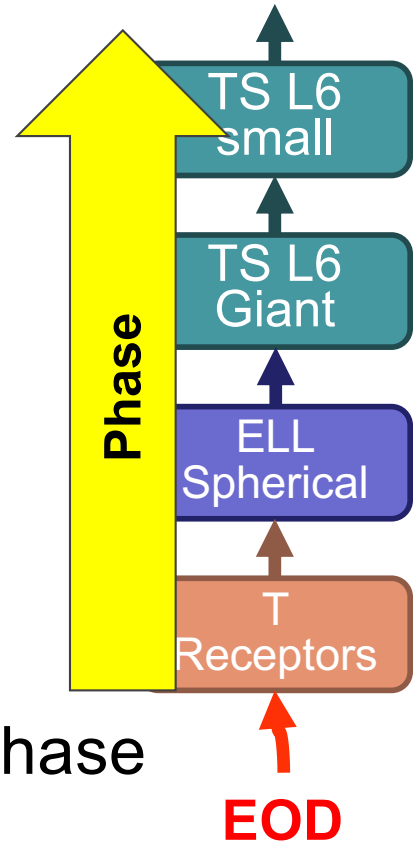




Phase coding

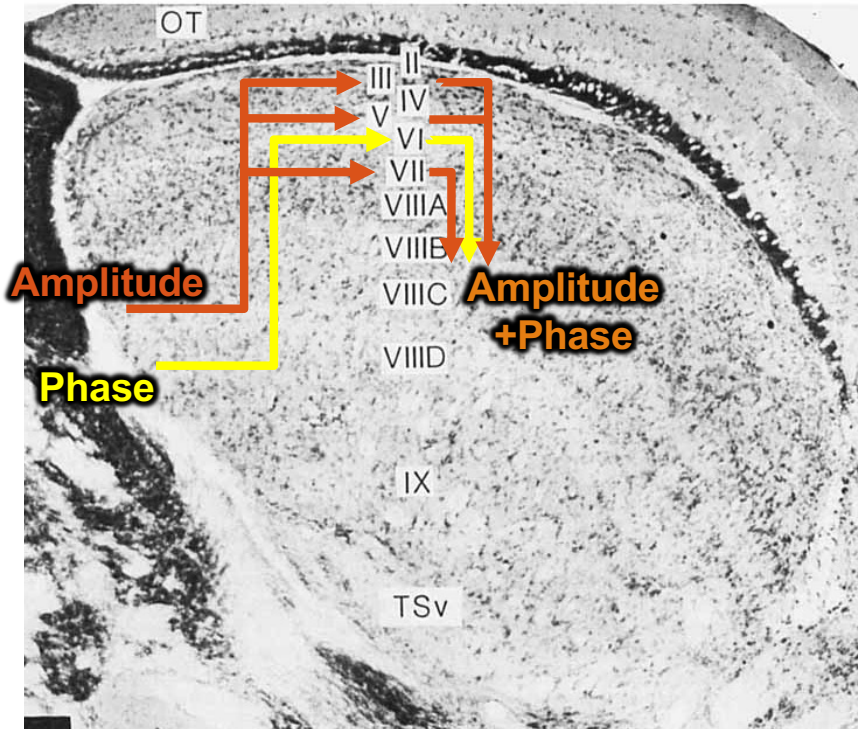
Split compartment experiments:
enable independent modulations of phase and amplitude

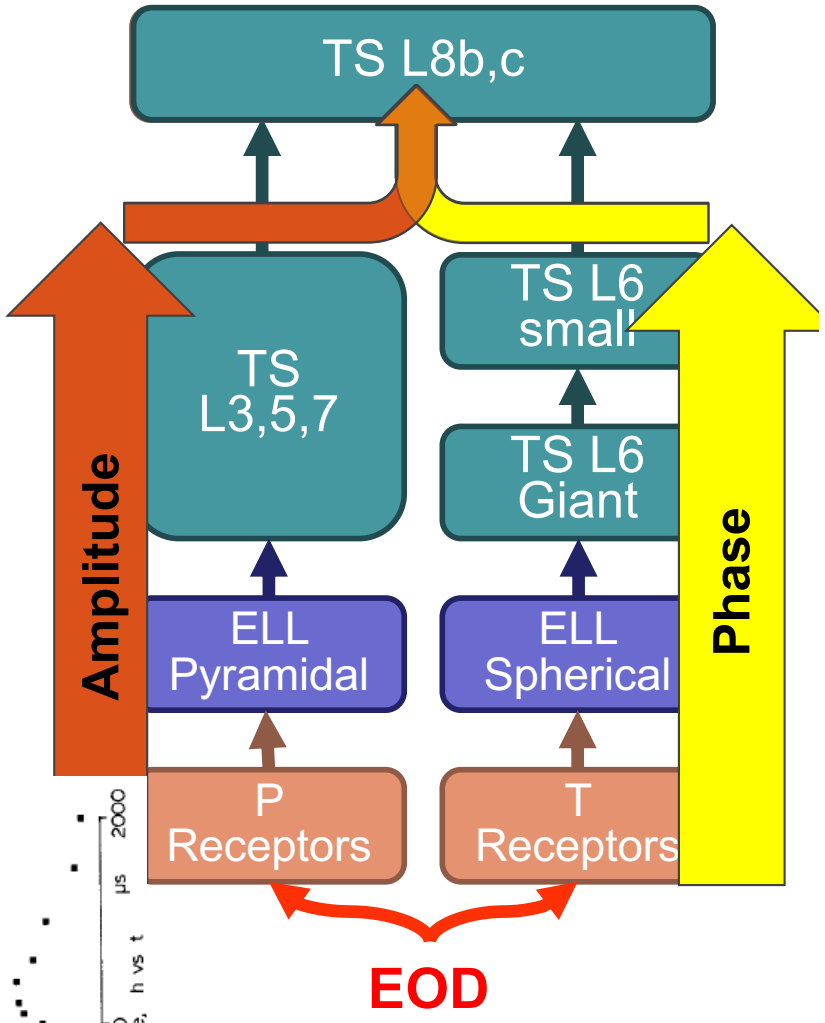


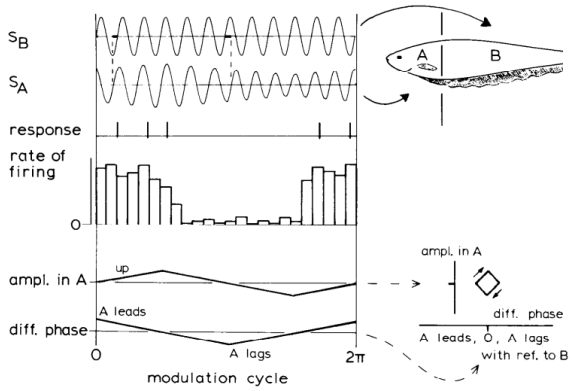


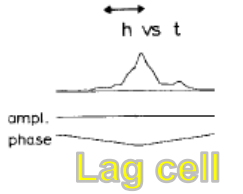
Small cells are coincidence detectors, reporting phase difference between two body regions.

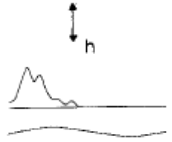
JAR physiology: Phase-amplitude coding

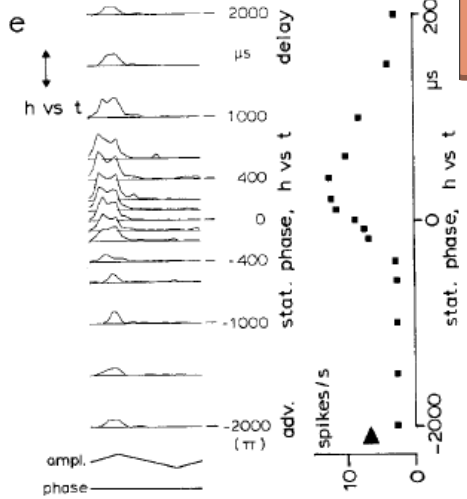








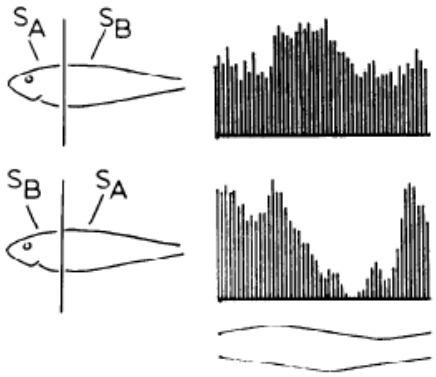






Phase-amplitude coding

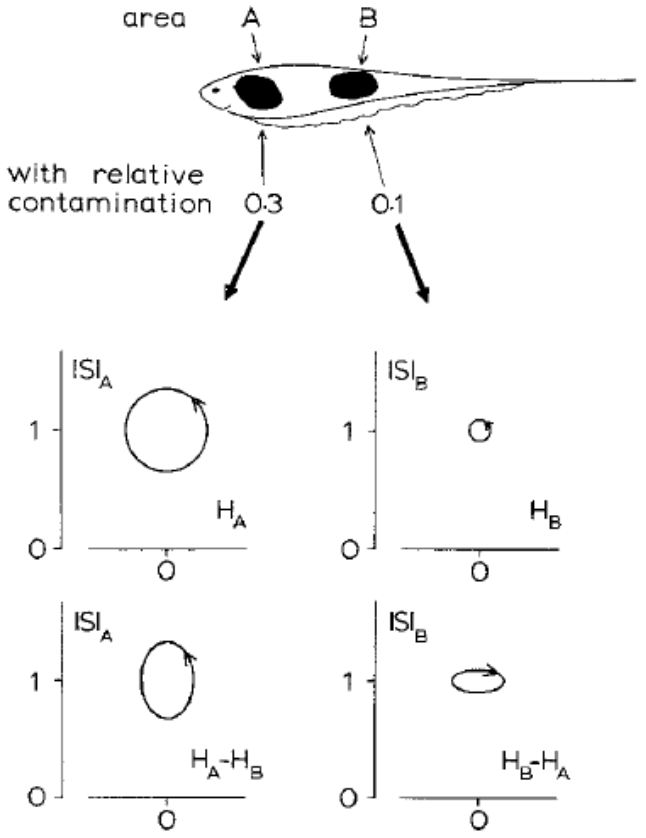
No Reference Signal- ambiguous code



E-Lead

E-Lag

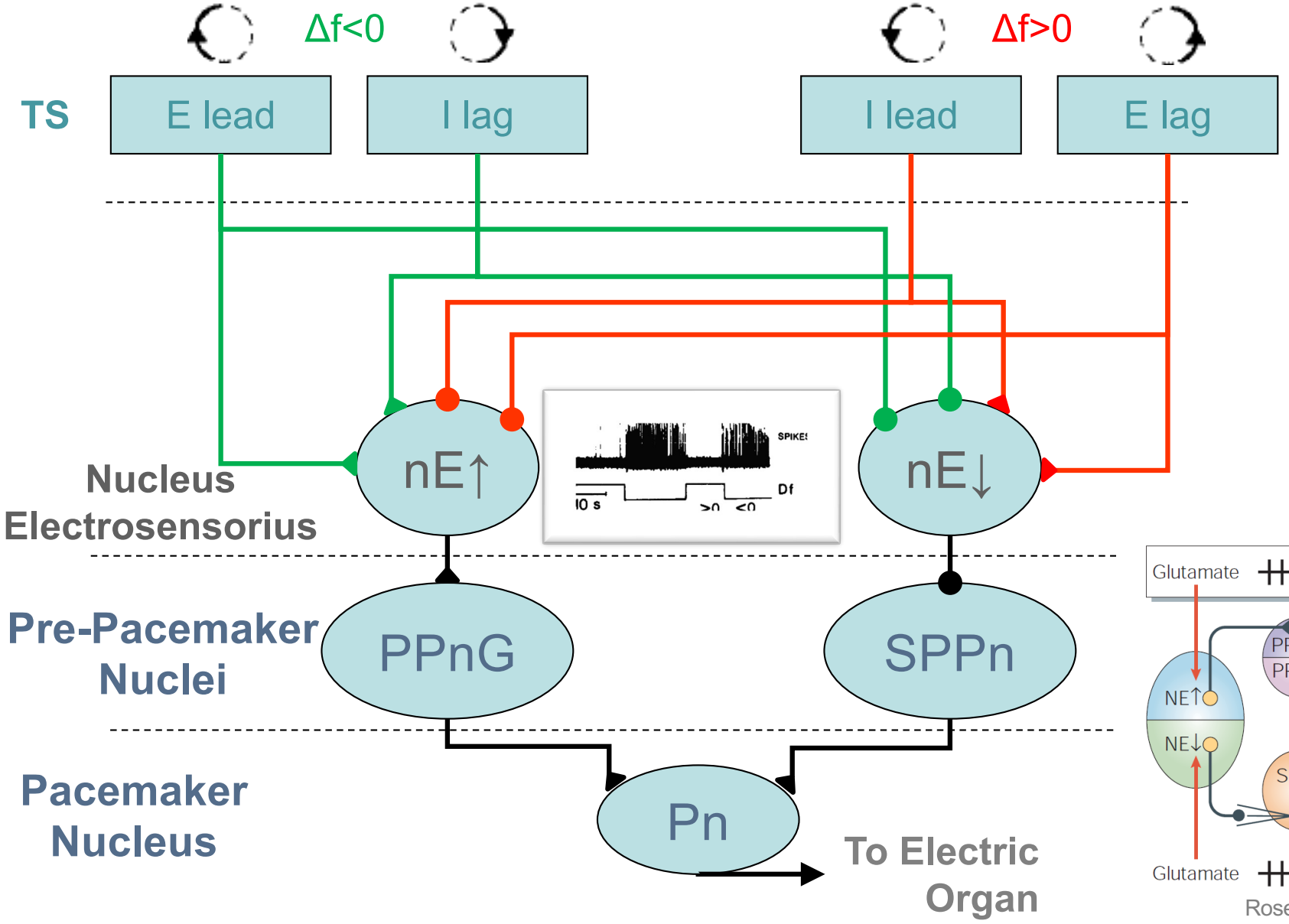
For each cell reporting Δf<0
there is a reciprocal cell reporting Δf>0

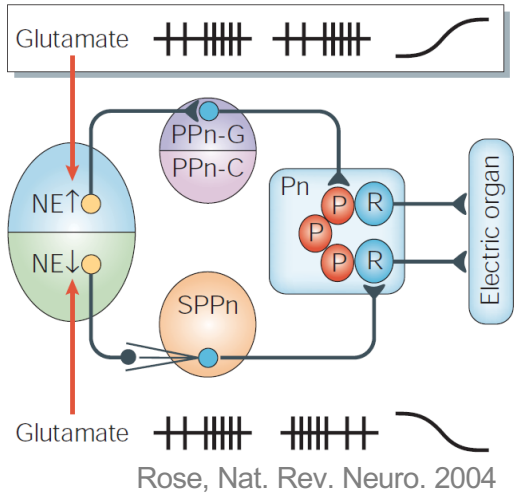


Solution: in natural geometry, different body regions are affected differently by the jamming interference.

Sensory-motor link

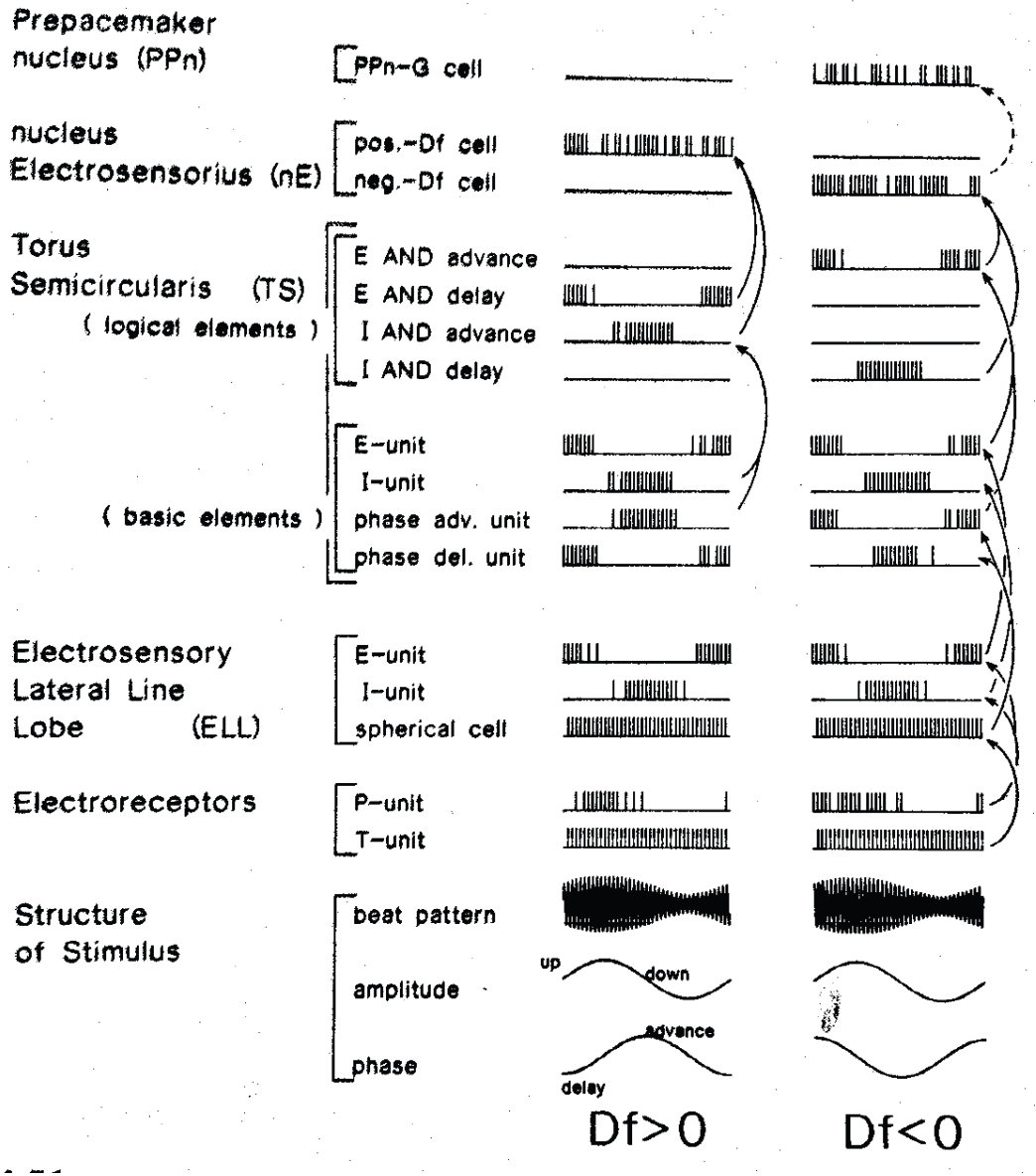
‘Neural Democracy’ (population code): the neurons ‘vote’ on the sign of Δf. The strongly affected regions get a stronger vote.

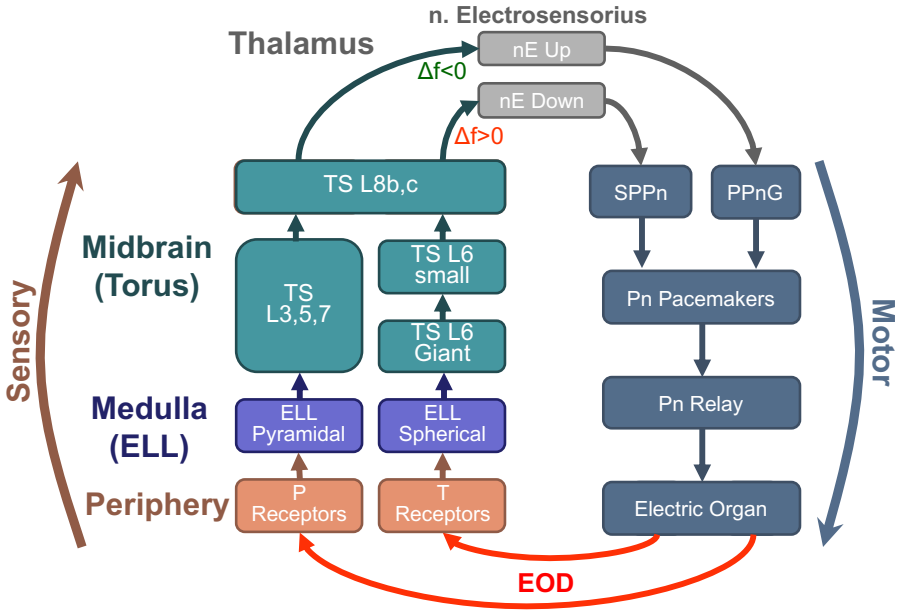


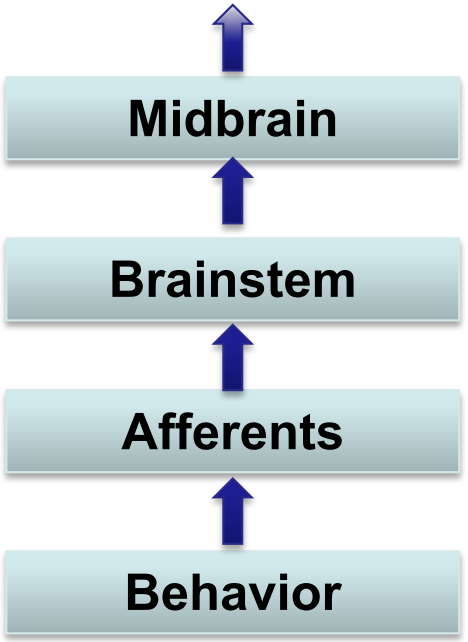


Rose, Nat. Rev. Neuro. 2004

JAR circuit- summary



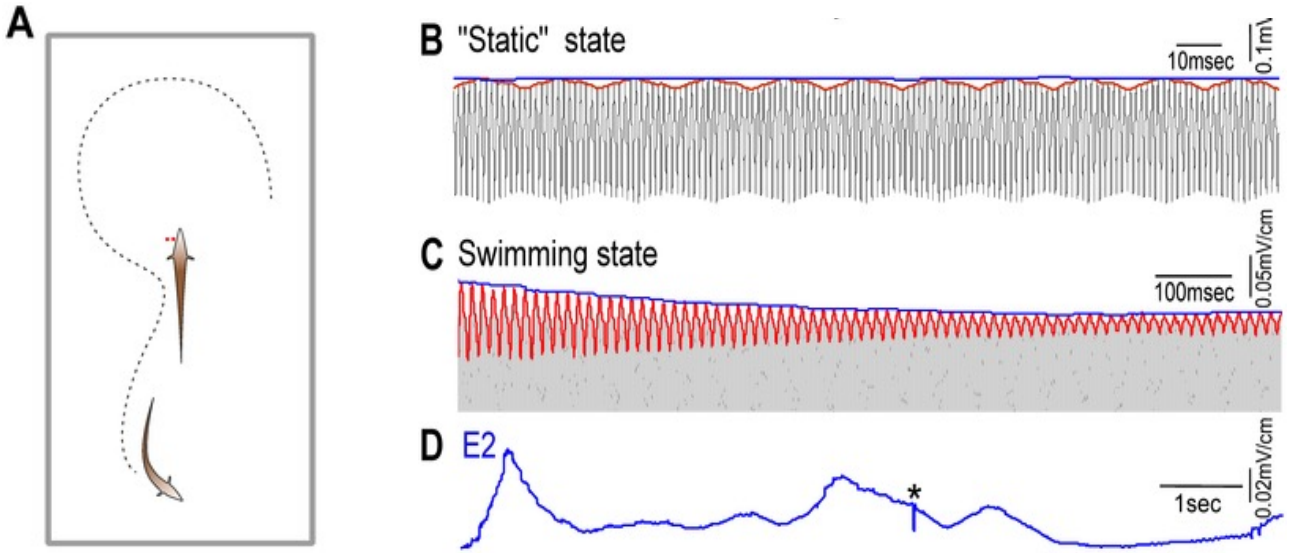




Chirps and Envelopes

The **beat frequency** depends on relative identity of other fish

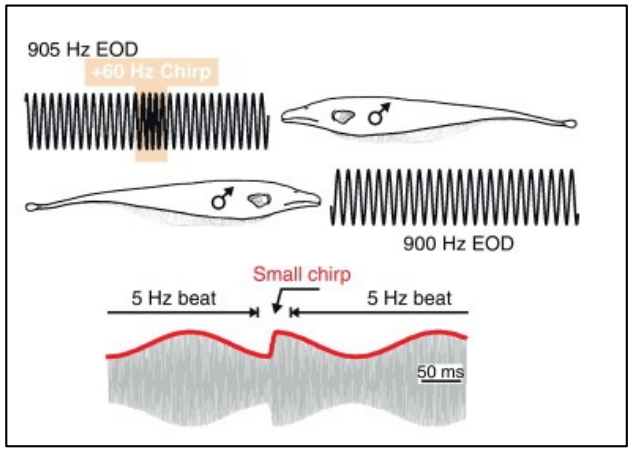
The '**beat envelope**' (modulation depth) depends on relative position of other fish



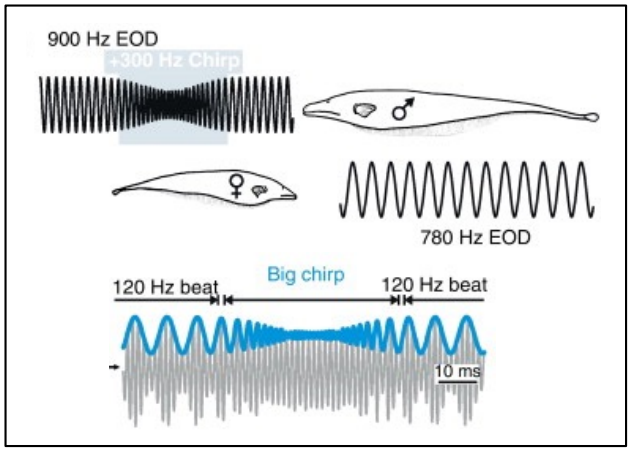
Yu et al, PLoS Comp. Biol. 2012

Chirps: transient changes in EOD frequency/amplitude, 'communication tokens'

Small chirps



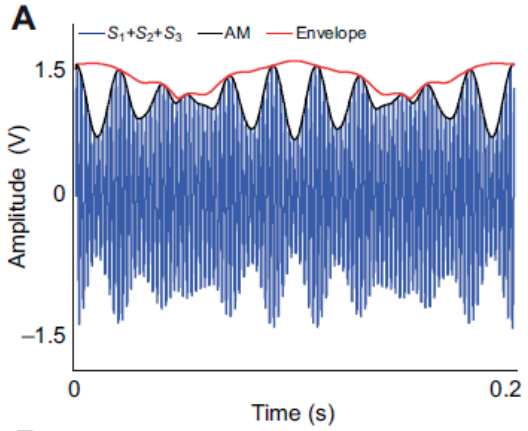
Long chirps

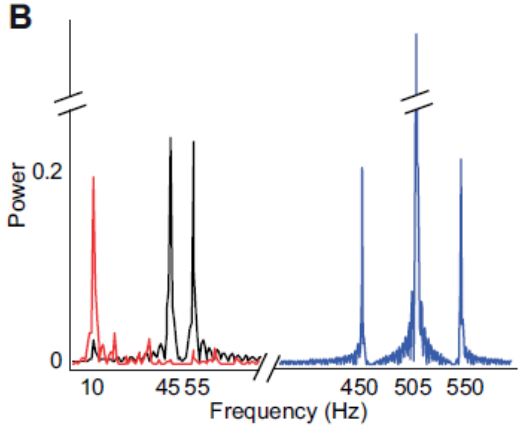


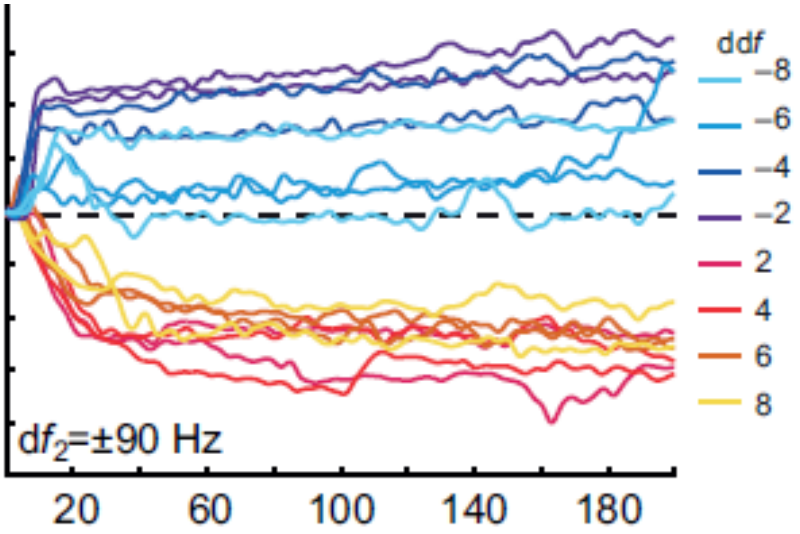
Marsat, Longtin & Maler, Curr. Op. Neurobiol. 2012

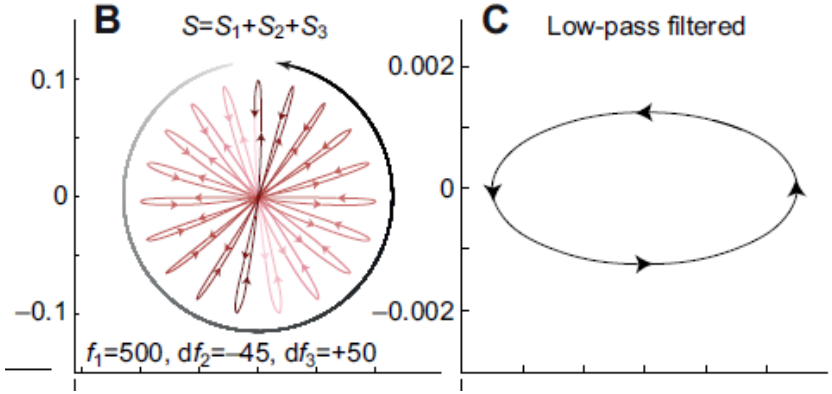
Social Envelope Response (SER)





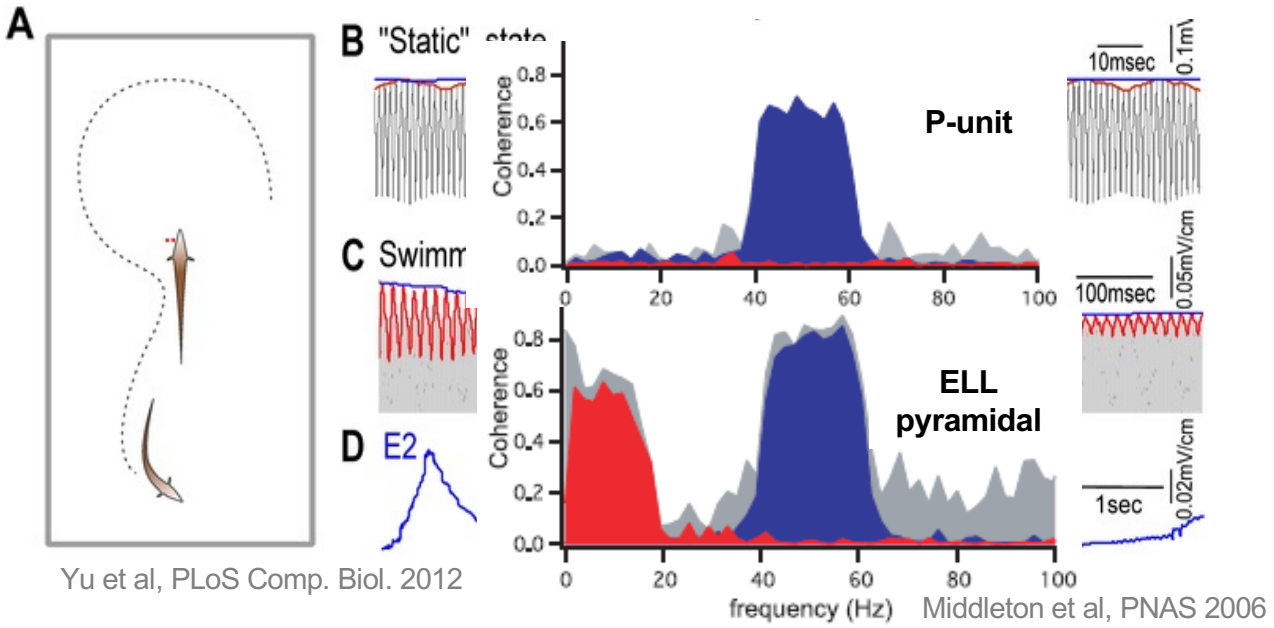






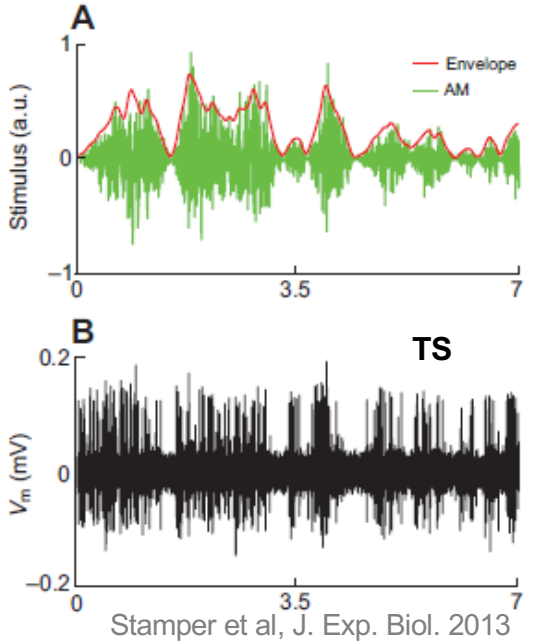
Stamper et al, J. Exp. Bio. 2012

Motion induced envelopes



Yu et al, PLoS Comp. Biol. 2012

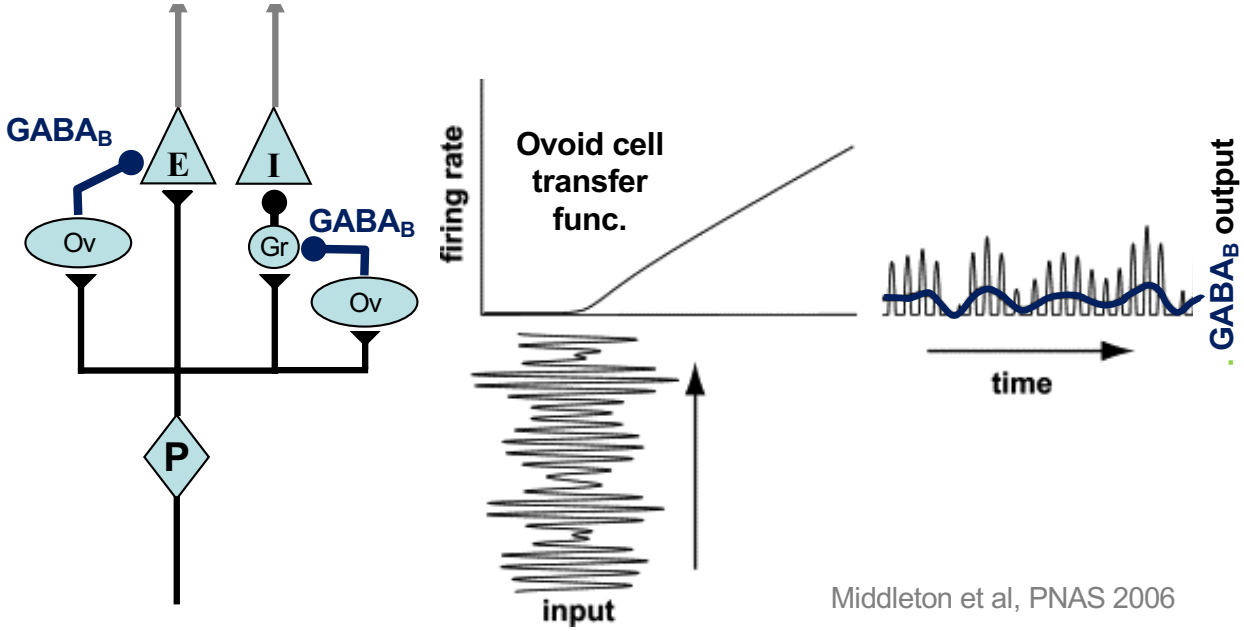
Middleton et al, PNAS 2006



Stamper et al, J. Exp. Biol. 2013

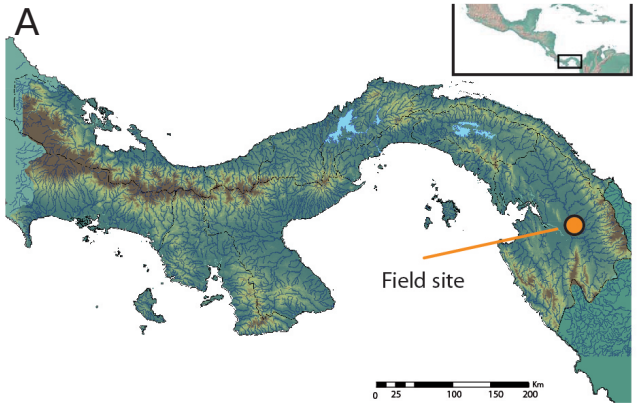
Q: How can pyramidal cells and TS cells encode envelopes when P-afferents don't?

A: Ovoid cells: high-freq. response, slow $GABA_B$ output.

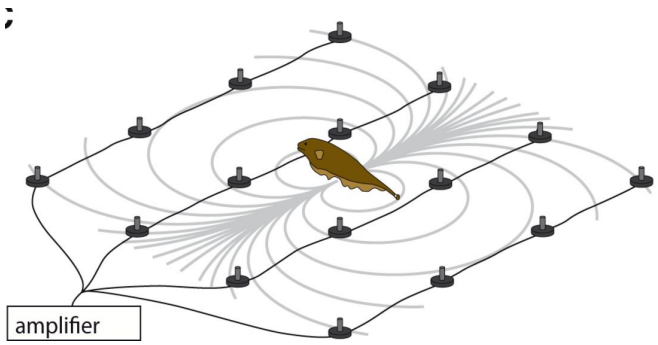


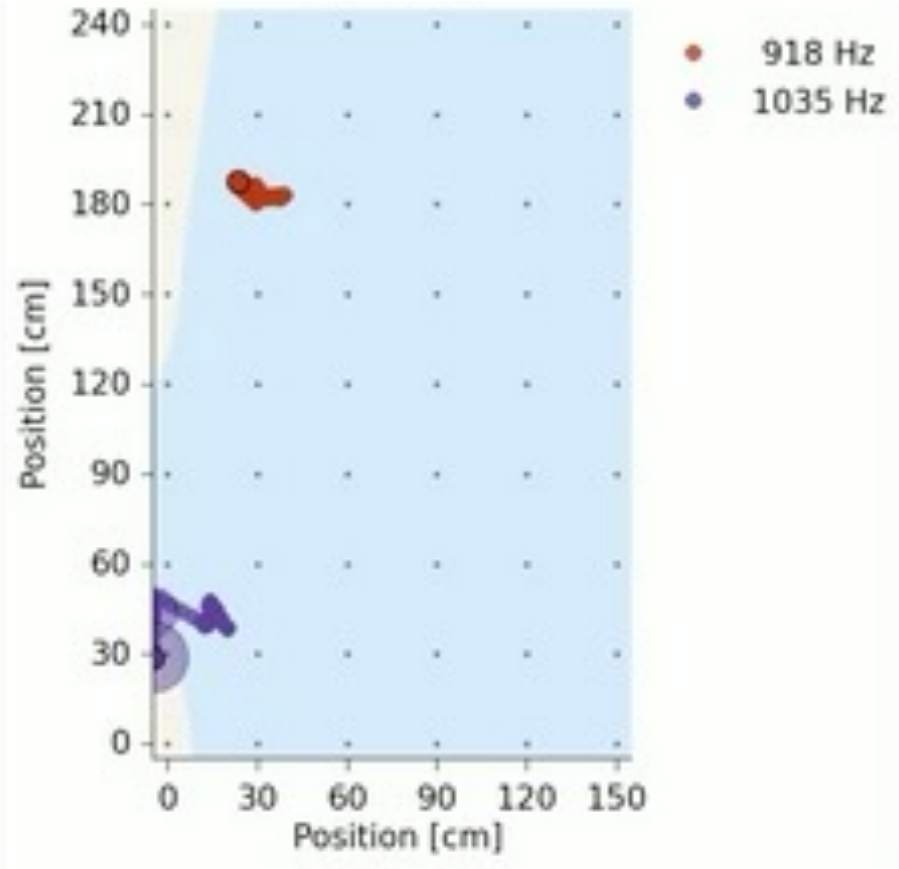
Middleton et al, PNAS 2006

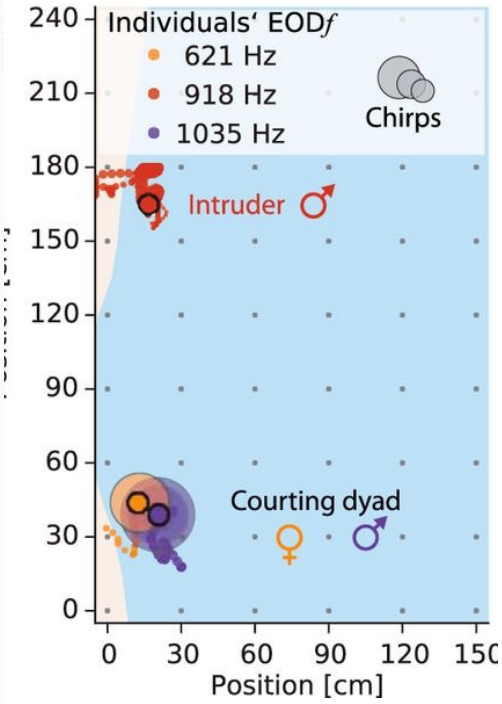
Electrocommunication in the wild



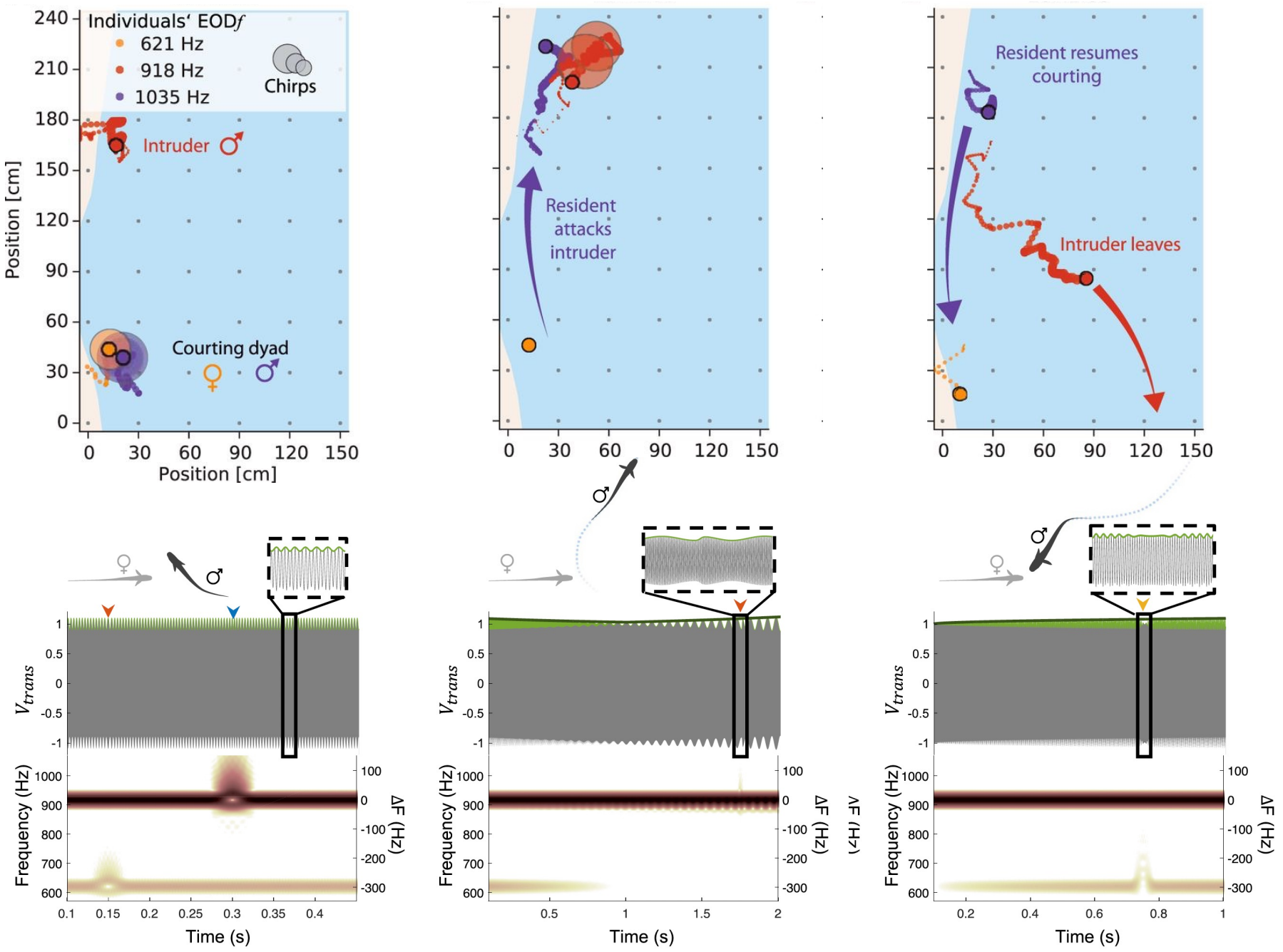








Henninger et al JNS 2018



Wallach et al, Curr. Bio. 2022

Henninger et al JNS 2018

Electrocommunication in the wild

Small chirps:

Opposing sex (high freq beat)➢courtship

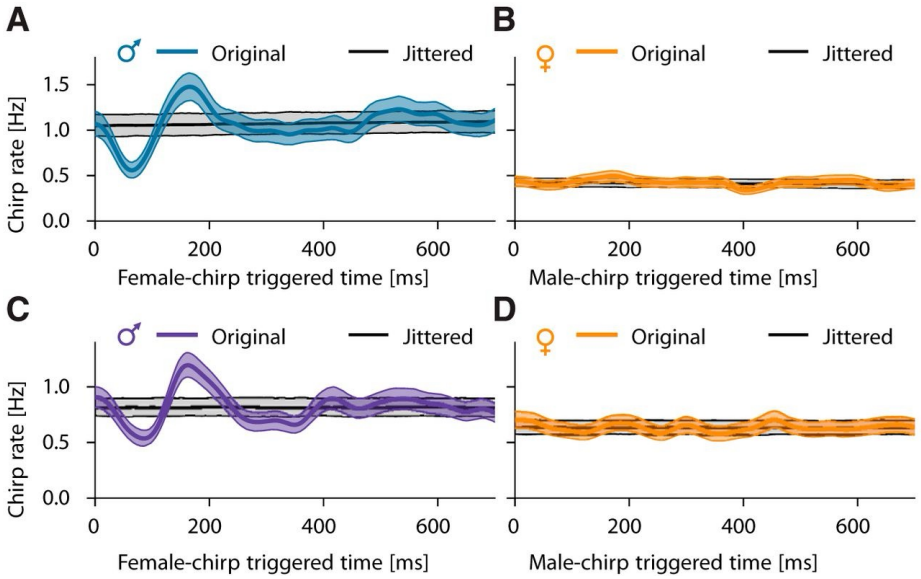
- Male echoes the female with precise latency



Same sex (low freq beat)➢submission

- Emitted by retreating intruder

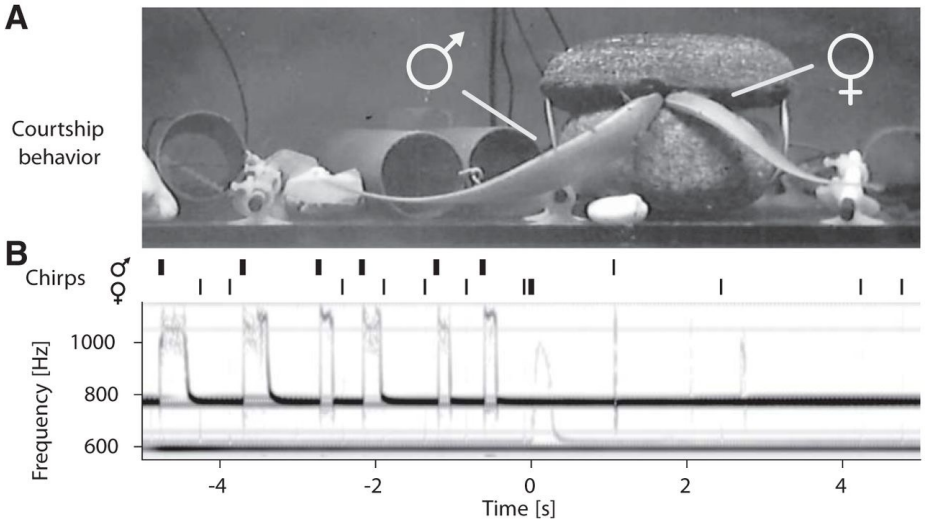




Long chirps:

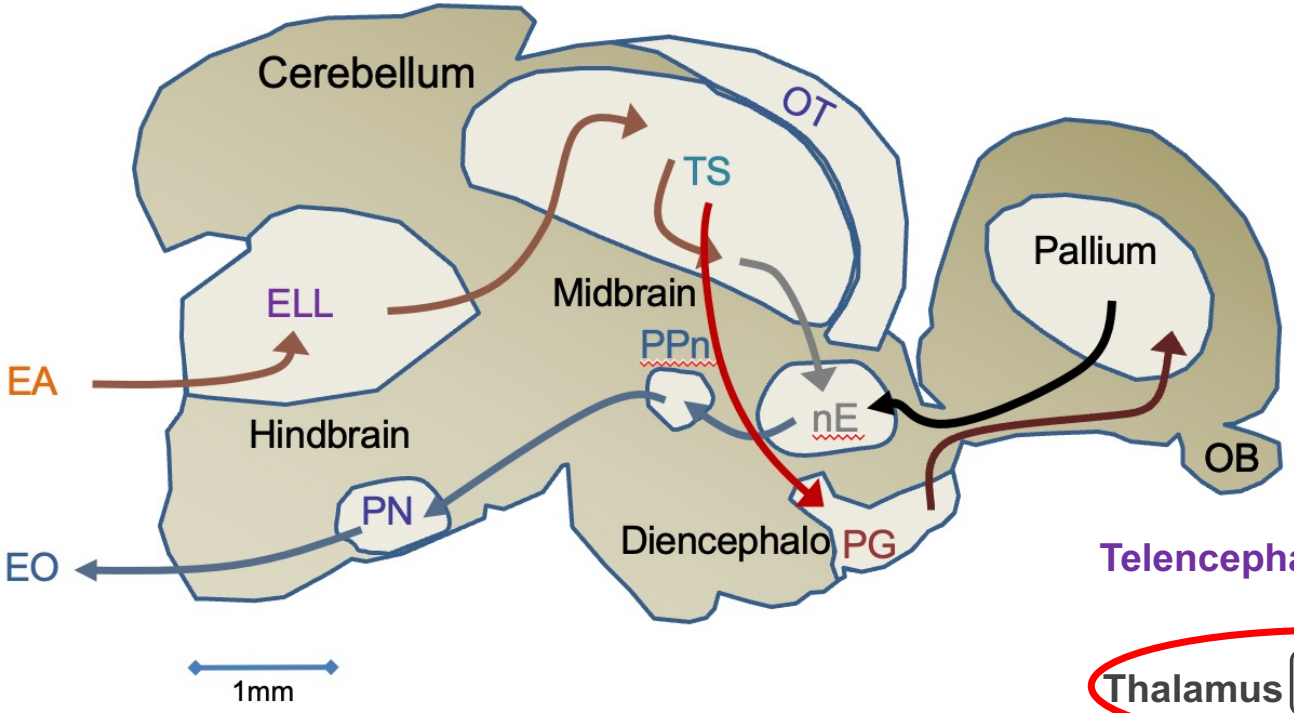
Courtship only ➢ female acceptance, spawning synchronization

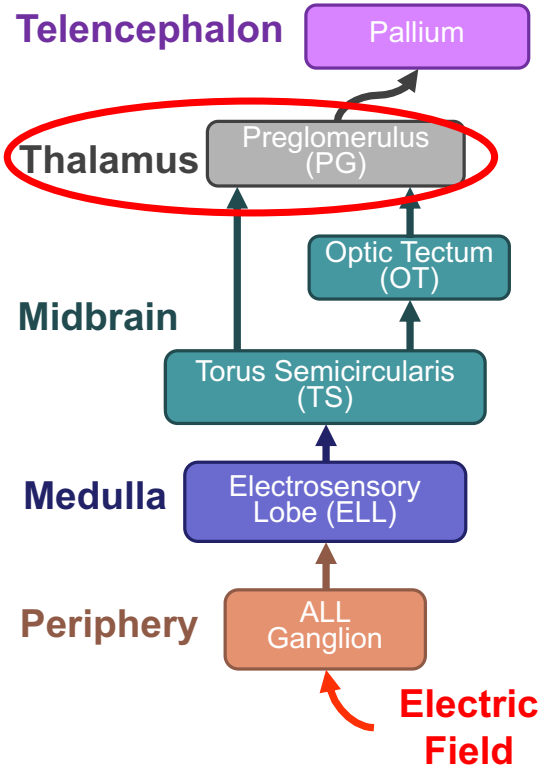




Henninger et al JNS 2018

Social Learning and Decision Making

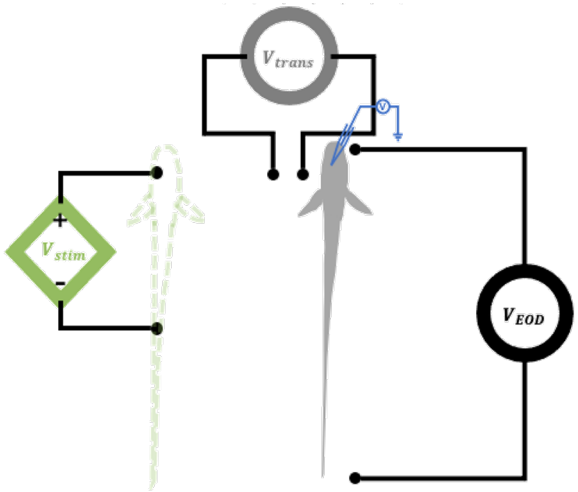


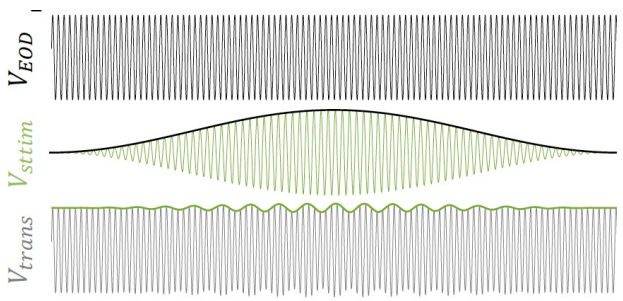


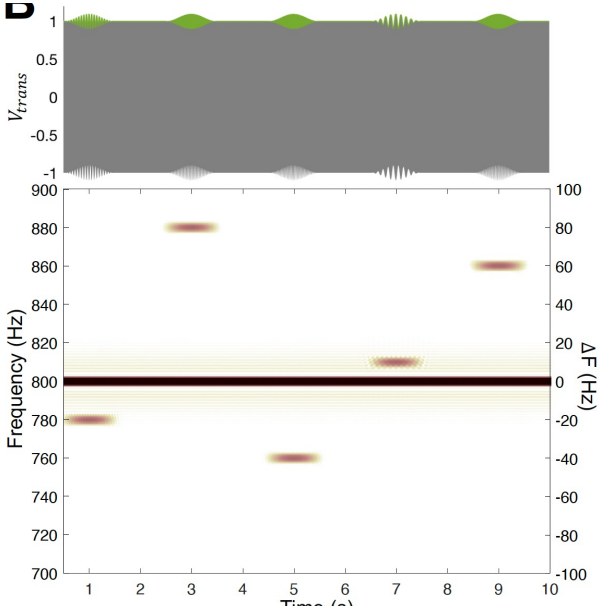
What Information is needed?

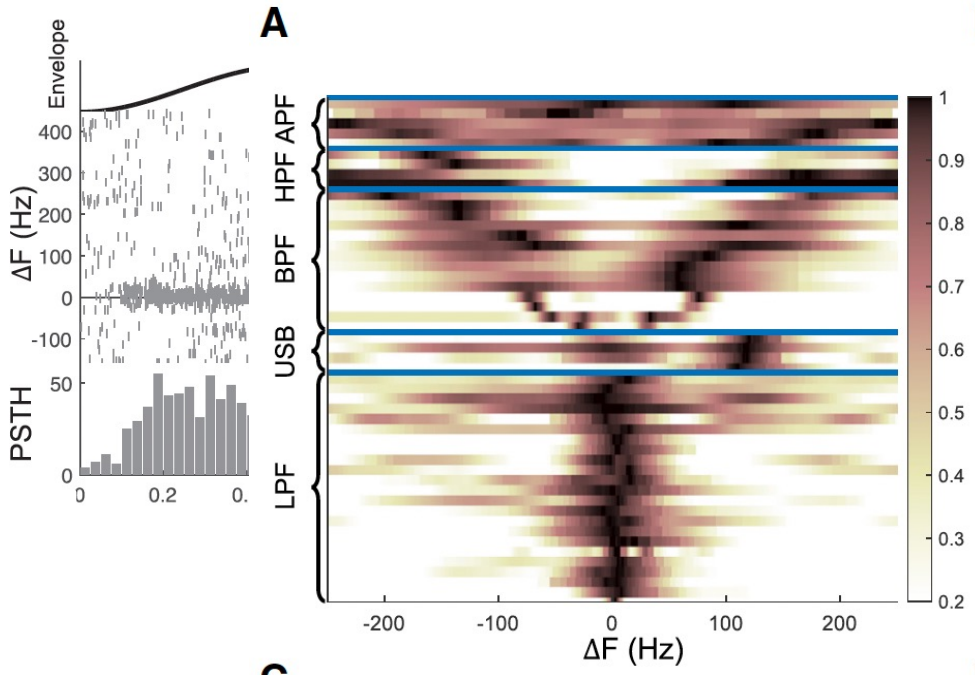
- Conspecific relative identity (Δf)
- Conspecific relative motion (envelope)
- Chirp type
- Chirp source
- Chirp timing

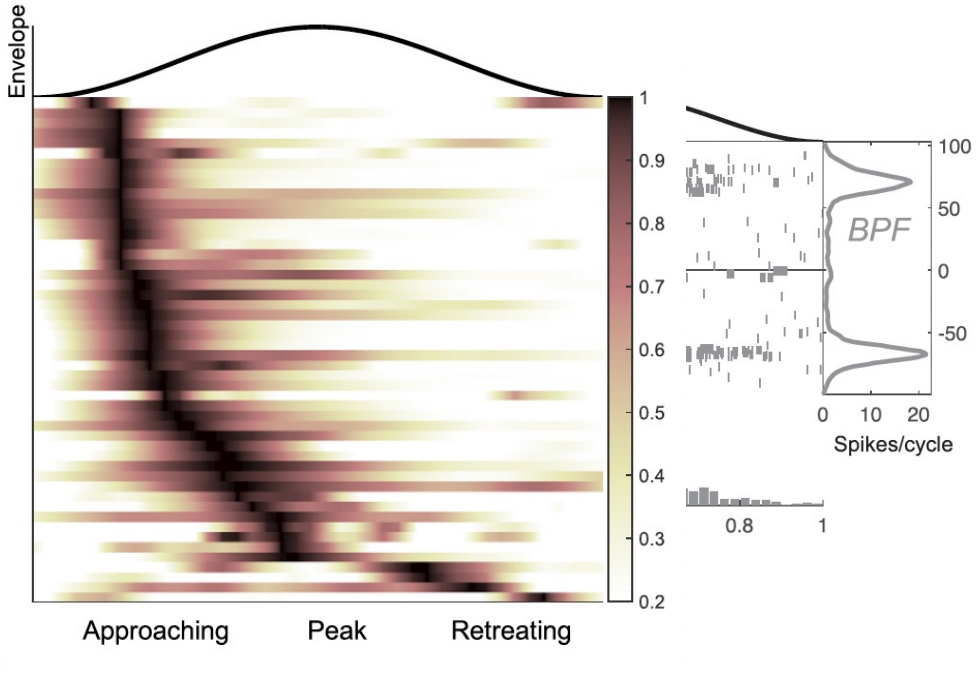
Thalamic Representation of Electrocommunication





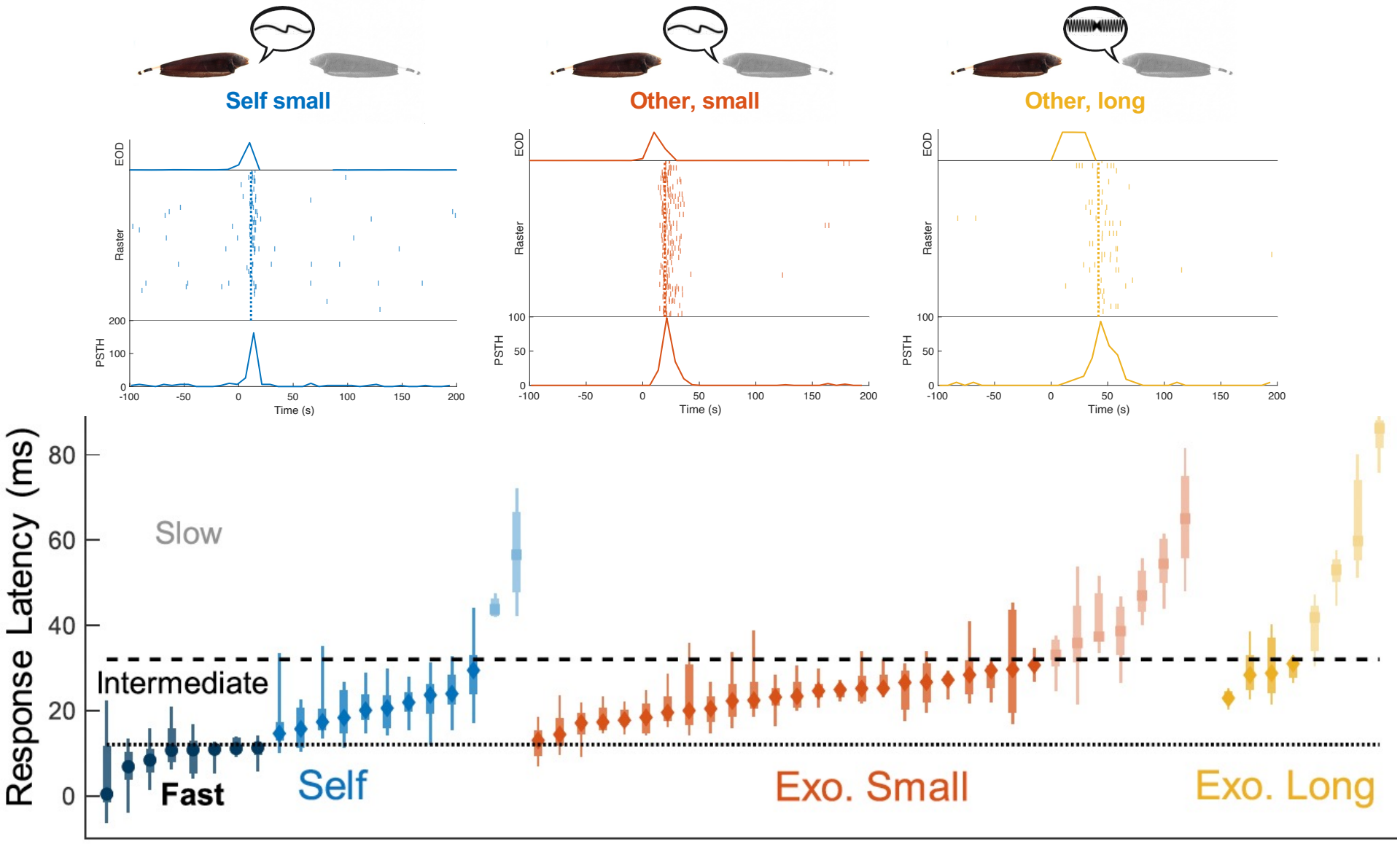






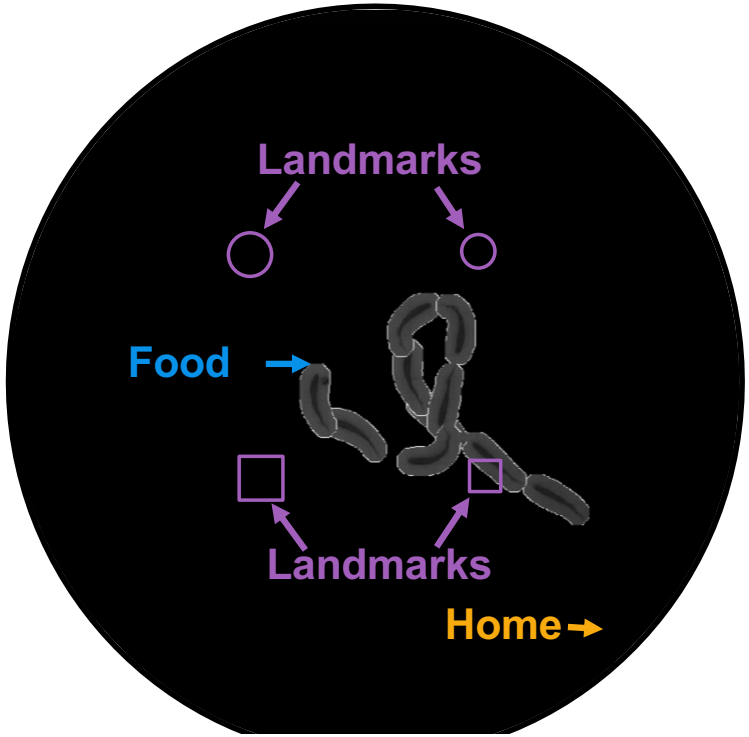
Wallach et al, Curr. Bio. 2022

Thalamic Representation of Electrocommunication

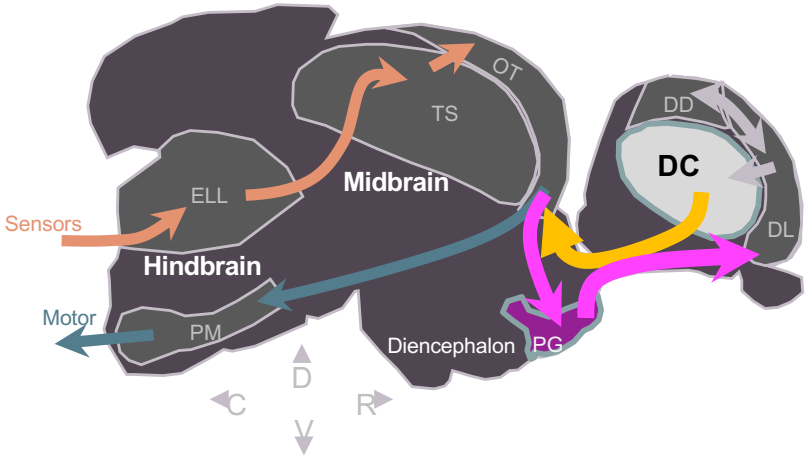


Wallach et al, Curr. Bio. 2022

Learning and Navigation



Jun et al, 2016



Pros and Cons of Electric Fish as an experimental system

Pros

- Convergent evolution
- Relatively few ethical constraints
- Low-dimensional behavior- easy to record and interpret
- Easy to mimic
- Easy to model
- Tractable neural circuitry
- Analogous to other modalities
- Highly-developed brain

Cons

- High-tech methods (imaging, genetics etc) are less-developed
- Captive breeding is challenging.
- Lives underwater…

SENSING, NAVIGATION, AND CONTROL IN LIVING SYSTEMS LAB

Recruiting students / postdocs!